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CGI Advantage 4

Instructor Guide

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Module 99: Comprehensive End-User Training Architecture, Strategy, and Plan for CGI Advantage 4 - FIN and HRM Modules



Overview

This document details the architecture, strategy, and plan for implementing end-user training for CGI Advantage 4's **Financial Management (FIN)** and **Human Resources Management (HRM)** modules. The goal is to equip end-users with the skills and knowledge required to perform their job functions effectively within the system. The training plan covers foundational knowledge, role-specific workflows, and practical application through hands-on exercises.

1. Training Architecture

The training architecture is structured around a blended learning model, combining **classroom training, e-learning modules, hands-on labs, and reference materials.**

Training Delivery Modes

Delivery Mode	Description
Classroom Training	Instructor-led sessions focusing on core concepts, system navigation, and basic functionalities. Ideal for onboarding and introducing key concepts.
E-Learning Modules	Self-paced online modules covering each function in detail. Allows users to revisit content as needed and provides flexibility for remote users.
Hands-On Labs	Practical exercises where users perform tasks in a sandbox environment. Enables application of knowledge in a low-risk setting.
Job Aids and Manuals	Reference materials including quick reference guides, job aids, and user manuals. Supports users in the workplace with quick look-up resources.

Training Curriculum Design

The curriculum is designed to align with core functional areas of the FIN and HRM modules, organized into **foundation training**, **role-specific training**, and **advanced feature training**:

- 1. Foundation Training:** Core functionalities, navigation, and common features across FIN and HRM modules.
- 2. Role-Specific Training:** Tailored to roles such as finance officers, HR managers, payroll clerks, and departmental administrators.
- 3. Advanced Features:** Covers complex functions such as budgeting adjustments, multi-entity financial reporting, payroll reconciliation, and benefit management.

2. Training Strategy

The training strategy is focused on maximizing user engagement, reinforcing learning through hands-on practice, and ensuring knowledge retention over time. This includes:

Objectives

- **Knowledge Transfer:** Ensure users understand system functionality and core processes.
- **Skill Development:** Build user confidence and proficiency in executing specific tasks.
- **Retention and Application:** Reinforce learning through job aids, practice, and ongoing support.

Key Elements

- 1. Targeted Needs Assessment:** Conduct a needs assessment to identify knowledge gaps, job-specific requirements, and user proficiency levels.
- 2. Curriculum Customization:** Tailor training modules based on assessment findings and align content with actual workflows and job requirements.
- 3. Evaluation and Feedback:** Implement pre- and post-assessments, quizzes, and surveys to measure user progress and adapt training as necessary.

Learning Paths by Role

Role	Learning Path
Finance Officer	Foundational FIN training, AP/AR functions, reporting, and financial reconciliation.
HR Manager	Foundational HRM training, employee records management, benefits administration, and reporting.
Payroll Clerk	Payroll setup, payroll processing, tax configuration, and payroll reconciliation.
Departmental Admin	Navigation, report generation, basic data entry, and cross-departmental workflows.

3. Training Plan

The training plan includes preparation, delivery, and post-training support to ensure successful adoption. Each phase is designed to build knowledge, reinforce skills, and provide continuous support for optimal system utilization.

Phase 1: Training Preparation

1. Develop Training Materials:

- Prepare e-learning modules, slides, job aids, and user manuals.
- Customize materials for CGI Advantage 4 FIN and HRM modules to reflect specific workflows and configurations.

2. Set Up Training Environment:

- Establish a sandbox or training environment in CGI Advantage 4 for hands-on exercises.
- Load realistic sample data for end-to-end testing of functions like invoice processing, payroll, and reporting.

3. Assign Training Schedules:

- Schedule sessions according to role and proficiency level.
- Notify users of their assigned training sessions and provide pre-training resources to set expectations.

Phase 2: Training Delivery

Module 1: System Navigation and Foundation Training

1. **Objective:** Familiarize users with CGI Advantage 4 interface, navigation, and general system functionality.
2. **Topics:**
 - Basic navigation: accessing modules, searching, and managing records.
 - Common fields, forms, and data entry standards across modules.
 - Overview of workflow and approval processes.

Module 2: Financial Management (FIN) Training

1. **Objective:** Train users on key financial functions within CGI Advantage 4.
2. **Topics:**
 - Accounts Payable: invoice entry, payment processing, and voucher reconciliation.
 - Accounts Receivable: billing, cash receipts, and customer account management.

- General Ledger: journal entries, trial balance, and financial statement generation.
- Budgeting: setting up budget structures, entering budgets, and monitoring budget controls.

3. **Hands-On Labs:** Practice entering invoices, creating journal entries, generating reports, and adjusting budgets.

Module 3: Human Resources Management (HRM) Training

1. **Objective:** Equip HR and payroll staff with skills for managing employee records, payroll, and benefits.
2. **Topics:**
 - Employee Profile: creating and updating employee records, managing personal and job data.
 - Payroll Processing: payroll run setup, tax configuration, and payroll calculation.
 - Benefits Management: enrolling employees, managing benefit plans, and handling benefit deductions.
 - Time and Attendance: entering time records, tracking leave balances, and leave request processing.
3. **Hands-On Labs:** Practice entering employee data, processing payroll, and managing benefits.

Module 4: Reporting and Advanced Functions

1. **Objective:** Teach users to generate, interpret, and analyze reports for informed decision-making.
2. **Topics:**
 - Standard Reports: financial statements, budget reports, payroll summaries, and HR reports.
 - Custom Reporting: using report tools to customize reports for specific needs.
 - Advanced Features: complex functions such as budget transfers, leave accruals, and benefits reconciliation.

Phase 3: Post-Training Support and Reinforcement

1. **Ongoing Support:**
 - Provide support through helpdesk resources, dedicated trainers, or designated "power users."
 - Offer drop-in sessions or "office hours" for users needing additional guidance.
2. **Job Aids and Manuals:**

- Distribute quick reference guides, troubleshooting checklists, and workflow summaries to all users.
- Make digital versions of all training materials accessible in an online repository.

3. Evaluation and Feedback:

- Conduct follow-up assessments to gauge knowledge retention.
- Collect user feedback on training effectiveness and areas for improvement.

4. Refresher Courses:

- Offer periodic refresher courses and updates as CGI Advantage 4 is upgraded or as new features are added.

4. Success Metrics and Evaluation

Evaluate training success based on user proficiency, system adoption, and feedback.

Key Metrics

- **User Proficiency:** Measured by post-training assessments and task completion rates.
- **System Adoption:** Observed through system usage metrics, such as the number of transactions processed.
- **Feedback Scores:** Gathered from surveys and user feedback, providing insight into training quality and areas for improvement.

Continuous Improvement

- Use evaluation data to continuously improve training materials and methods.
- Update training as CGI Advantage 4 features or modules are enhanced, ensuring training remains relevant and effective.

Summary

This training architecture, strategy, and plan are designed to enable end-users to leverage the full functionality of CGI Advantage 4's FIN and HRM modules. Through structured learning paths, hands-on practice, and ongoing support, users will develop the skills necessary for efficient, accurate, and effective system utilization.

Introduction to Mastering CGI Advantage 4

Welcome to the course **Mastering CGI Advantage 4**. This comprehensive course is designed to equip you with the knowledge and skills necessary to effectively navigate and utilize CGI Advantage 4, a leading enterprise resource planning (ERP) solution tailored for government agencies and public sector organizations.

Course Overview

CGI Advantage 4 is a powerful, integrated ERP system that supports a wide range of functions, including financial management, procurement, human resources, and more. This course will take you on a journey through the various modules and features of CGI Advantage 4, providing you with the practical knowledge needed to manage your organization's operations efficiently.

Throughout this course, you will engage in hands-on exercises, explore real-world case studies, and learn best practices for implementing and maintaining CGI Advantage 4 in a public sector environment. By the end of the course, you will be well-prepared to apply these skills in your role, whether you are a system administrator, financial manager, procurement specialist, or HR professional.

Who Should Take This Course?

This course is ideal for:

- Public sector professionals who are responsible for managing or overseeing ERP systems.
- System administrators and IT professionals tasked with implementing and maintaining CGI Advantage 4.
- Financial managers, procurement officers, and HR professionals who use CGI Advantage 4 in their daily operations.
- Anyone preparing for certification in CGI Advantage 4 or seeking to enhance their ERP expertise.

Learning Objectives

By the end of this course, you will be able to:

- Understand the architecture and components of CGI Advantage 4.
- Configure and manage the core functionalities, including financial management, procurement, and HR modules.
- Implement and optimize advanced features such as workflow automation, reporting, and system integration.
- Apply best practices for system administration, security, and compliance.
- Prepare for CGI Advantage 4 certification through practical exercises and exam preparation tips.
- Complete a final project that demonstrates your mastery of the system.

Course Structure

The course is divided into nine modules, each focusing on a specific aspect of CGI Advantage 4:

1. Introduction to CGI Advantage 4
2. System Setup and Configuration
3. Core Functionalities
4. Advanced Functionalities
5. System Administration and Maintenance
6. Reporting and Analytics
7. Practical Applications and Case Studies
8. Best Practices and Future Directions
9. Certification and Final Project

Each module includes lessons and topics that delve into the details of the system, with a mix of theoretical knowledge and practical exercises. You'll have the opportunity to apply what you learn through hands-on activities and case studies that mirror real-world scenarios.

Getting Started

Before you begin, ensure you have access to a CGI Advantage 4 environment for practice and hands-on exercises. This will allow you to apply the concepts discussed in each module directly within the system, reinforcing your learning and building your confidence.

We encourage you to actively participate in all exercises, take notes, and engage with any additional resources provided throughout the course. Your success in this course will be measured not only by your understanding of the material but also by your ability to apply what you've learned in practical settings.

Thank you for enrolling in **Mastering CGI Advantage 4**. Let's get started on your journey to becoming an expert in this powerful ERP system!

Module 01: Introduction to CGI Advantage 4



Learning Objectives



Upon completion of this module, students should be able to:

- Understand the basic concepts and functionalities of CGI Advantage 4.
- Gain an overview of the system architecture and core components.
- Learn about the history and evolution of CGI Advantage 4, including key features and * benefits.

Overview of CGI Advantage

Introduction to Enterprise Resource Planning (ERP)

Enterprise Resource Planning (ERP) systems are integrated software platforms that manage core business processes, including finance, procurement, HR, and supply chain operations. ERP systems centralize data and processes, allowing organizations to operate more efficiently by improving communication, streamlining processes, and providing real-time insights.

- **Key Benefits of ERP Systems:**

- Integrated Business Processes
- Improved Data Accuracy
- Enhanced Reporting and Analysis
- Better Decision-Making Support

History and Evolution of CGI Advantage

CGI Advantage is a comprehensive ERP solution designed specifically for government agencies and public sector organizations. Originally developed by CGI Group, the software has evolved over decades, incorporating advancements in technology and addressing the unique needs of the public sector.

- **Milestones in CGI Advantage History:**

- 1990s: Initial release of CGI Advantage
- 2000s: Introduction of web-based interfaces and integration capabilities
- 2010s: Enhanced modular design and cloud deployment options
- 2020s: Release of CGI Advantage 4 with modernized architecture and advanced features

Key Features and Benefits of CGI Advantage 4

CGI Advantage 4 is the latest version of CGI's ERP solution, offering a robust and flexible platform tailored for public sector organizations. Its features are designed to improve operational efficiency, enhance service delivery, and ensure compliance with government regulations.

- **Key Features of CGI Advantage 4:**

- **Modular Design:** Easily customizable modules for finance, procurement, HR, and more.
- **Cloud Deployment:** Scalable cloud options to reduce infrastructure costs and improve accessibility.
- **Enhanced Security:** Advanced security measures to protect sensitive government data.
- **Reporting and Analytics:** Powerful tools for real-time reporting and data-driven decision-making.

System Architecture and Components

Understanding the System Architecture

The architecture of CGI Advantage 4 is designed to be modular, scalable, and flexible, enabling public sector organizations to implement the software according to their specific needs. The system is built on a service-oriented architecture (SOA) that supports integration with other systems and allows for easy updates and enhancements.

- **Core Components of the Architecture:**

- Application Layer: Handles business logic and operations.
- Data Layer: Manages data storage, retrieval, and processing.
- Integration Layer: Facilitates communication between different modules and external systems.

Overview of Core Components and Modules

CGI Advantage 4 is composed of several core modules that cover a wide range of functions necessary for public sector operations. These modules can be implemented individually or as part of an integrated system, depending on the organization's needs.

- **Core Modules:**

- Financial Management: General Ledger, Accounts Payable/Receivable, Budgeting.
- Procurement Management: Vendor Management, Contract Management, Purchase Orders.
- Human Resources and Payroll: Employee Data Management, Payroll Processing, Benefits Administration.
- Workflow Automation: Automated processes for approvals, notifications, and task management.

Integration with Other Systems

One of the key strengths of CGI Advantage 4 is its ability to integrate with other systems, both within and outside the organization. This capability is crucial for ensuring seamless operations and data consistency across different departments and external partners.

- **Integration Capabilities:**

- **API Integration:** Allows for custom integrations with third-party software.
- **Data Exchange:** Supports standard data formats for easy exchange between systems.
- **Real-Time Syncing:** Ensures that data is updated in real-time across integrated systems.

Module : Advanced Functionalities



Learning Objectives



Upon completion of this module, students should be able to:

- Gain expertise in fund accounting, asset management, and advanced reporting within CGI Advantage 4.
- Learn how to design, implement, and manage workflows to automate business processes.
- Understand how to integrate CGI Advantage 4 with external systems, manage APIs, and set up data exchange processes.

Advanced Financial Features

Fund Accounting

Fund accounting is a specialized accounting method used by government and nonprofit organizations to track the use of funds, ensuring that resources are used according to donor or grantor stipulations. CGI Advantage 4 provides robust support for fund accounting, allowing organizations to manage multiple funds within the same accounting framework.

- **Setting Up Funds:**

- Define fund structures, including restrictions and usage guidelines.
- Assign funds to specific programs, projects, or departments.

- **Managing Fund Balances:**

- Track fund balances in real-time.
- Generate reports on fund utilization and remaining balances.

- **Fund Reporting:**

- Create customized fund reports to meet external audit requirements.
- Ensure compliance with donor or grantor stipulations.

Asset Management

Asset management involves tracking, maintaining, and managing the organization's assets, from acquisition to disposal. CGI Advantage 4 includes a comprehensive asset management module that integrates with financial management and procurement systems.

- **Asset Inventory:**

- Record and categorize assets, including fixed assets, equipment, and vehicles.
- Track asset locations, conditions, and depreciation.

- **Asset Lifecycle Management:**

- Manage the acquisition, maintenance, and disposal of assets.
- Schedule and track asset maintenance activities.

- **Asset Reporting:**

- Generate depreciation schedules and financial reports on asset value.
- Monitor asset utilization and optimize asset performance.

Advanced Reporting and Analytics

Advanced reporting and analytics in CGI Advantage 4 enable organizations to gain deeper insights into their operations, support data-driven decision-making, and meet complex reporting requirements. This topic covers the use of advanced tools for creating custom reports and performing data analysis.

- **Custom Report Creation:**

- Use the report designer tool to create custom financial, procurement, and HR reports.
- Implement filters, parameters, and calculations to tailor reports to specific needs.

- **Data Analytics:**

- Perform trend analysis and forecasting using historical data.
- Implement predictive analytics to anticipate future financial and operational outcomes.

- **Data Visualization:**

- Create dashboards with visualizations, such as charts and graphs, to present data clearly.
- Share dashboards with stakeholders for real-time insights.

Workflow Automation

Introduction to Workflow Automation

Workflow automation streamlines business processes by automating repetitive tasks and ensuring consistent execution. CGI Advantage 4 includes a powerful workflow engine that automates processes across financial management, procurement, HR, and other modules.

- **Benefits of Workflow Automation:**

- Increase efficiency by reducing manual intervention.
- Improve accuracy and compliance by standardizing processes.
- Enhance transparency by providing audit trails for automated workflows.

- **Workflow Automation Examples:**

- Automate approval processes for purchase orders, contracts, and budget adjustments.
- Implement automated notifications and reminders for pending tasks.

Creating and Managing Workflows

Creating and managing workflows in CGI Advantage 4 involves defining the steps, conditions, and actions that make up a business process. This topic covers the steps to design, implement, and maintain workflows.

- **Designing Workflows:**

- Use the workflow designer tool to map out processes visually.
- Define triggers, conditions, and actions for each step of the workflow.

- **Implementing Workflows:**

- Deploy workflows within the system and assign tasks to users or roles.
- Test workflows in a staging environment before going live.

- **Managing and Monitoring Workflows:**

- Monitor workflow performance and identify bottlenecks or delays.
- Modify workflows as needed to accommodate changes in business processes.

Monitoring and Optimizing Workflows

Monitoring and optimizing workflows is essential to ensure that automated processes remain efficient and effective over time. This topic covers the tools and techniques for tracking workflow performance and making adjustments to improve outcomes.

- **Workflow Monitoring Tools:**

- Use built-in analytics to track workflow performance metrics, such as completion times and error rates.
- Set up alerts for workflow issues, such as overdue tasks or failed automation steps.

- **Optimizing Workflow Performance:**

- Analyze workflow data to identify inefficiencies or areas for improvement.
- Implement changes to workflow steps, conditions, or automation rules to enhance performance.

- **Continuous Improvement:**

- Regularly review and refine workflows based on user feedback and changing business needs.
- Stay up-to-date with software updates that may introduce new workflow automation features.

System Integration

Integration with External Systems

CGI Advantage 4 supports integration with a wide range of external systems, allowing organizations to connect their ERP system with other software used for specialized functions, such as tax processing, HR systems, or third-party financial applications.

- **Integration Methods:**

- API Integration: Use REST or SOAP APIs to connect CGI Advantage 4 with external systems.
- Batch Data Integration: Exchange data through batch processes using standard formats (e.g., CSV, XML).

- **Key Integration Use Cases:**

- Integrate with state or federal tax systems for automatic tax calculations and reporting.
- Connect with external HR systems for seamless employee data exchange.

- **Ensuring Data Consistency:**

- Implement data validation and error handling during integration to maintain data integrity.
- Schedule regular data synchronization to keep systems up-to-date.

API Management

API management is critical for ensuring secure and efficient communication between CGI Advantage 4 and external systems. This topic covers best practices for managing APIs, including authentication, rate limiting, and monitoring.

- **API Security:**

- Implement API authentication methods, such as OAuth or API keys, to secure endpoints.
- Use encryption (e.g., HTTPS) to protect data in transit.

- **API Rate Limiting:**

- Set up rate limits to control the number of API requests made within a specific timeframe.
- Monitor API usage to prevent abuse and ensure system performance.

- **API Monitoring and Troubleshooting:**

- Use logging and monitoring tools to track API calls and identify issues.
- Implement automated alerts for API errors or performance degradation.

Data Exchange and ETL Processes

Data exchange and Extract, Transform, Load (ETL) processes are essential for moving data between CGI Advantage 4 and other systems. This topic covers the setup and management of ETL processes to ensure accurate and efficient data transfers.

- **ETL Process Setup:**

- Define data sources and destinations, including databases, flat files, or external APIs.
- Design data transformation rules to format and clean data before loading.

- **Managing Data Transfers:**

- Schedule regular ETL jobs to automate data transfers between systems.
- Monitor ETL processes to ensure data is transferred correctly and efficiently.

- **Data Validation and Error Handling:**

- Implement data validation rules to catch errors during the ETL process.
- Set up error handling procedures, such as retry mechanisms or alerts, to address issues promptly.

Module 24: System Administration and Maintenance



Learning Objectives



Upon completion of this module, students should be able to:

- Develop skills in user account management, including the setup of roles, permissions, and authentication.
- Learn best practices for maintaining system security and compliance with regulatory standards.
- Understand how to monitor system performance, troubleshoot common issues, and optimize performance for efficiency.

System Administration

User Account Management

User account management is a critical aspect of system administration, ensuring that only authorized personnel have access to the system. CGI Advantage 4 provides robust tools for managing user accounts, roles, and permissions.

- **Creating User Accounts:**
 - Set up new user accounts with appropriate access rights.
 - Define user roles based on job responsibilities.
- **Managing User Roles:**
 - Assign roles to users, determining their access to different modules and features.
 - Modify or revoke roles as needed to reflect changes in job duties.
- **User Authentication:**
 - Implement multi-factor authentication (MFA) for added security.
 - Manage password policies, including complexity requirements and expiration periods.

Security and Compliance

Ensuring the security of the CGI Advantage 4 system is paramount, especially given the sensitive nature of the data managed by public sector organizations. This topic covers best practices for securing the system and ensuring compliance with relevant regulations.

- **System Security Measures:**

- Implement role-based access control (RBAC) to limit user access based on roles.
- Use encryption for data at rest and in transit to protect sensitive information.

- **Compliance Management:**

- Ensure the system meets government regulations, such as HIPAA, GDPR, or FISMA.
- Generate compliance reports for audits and regulatory reviews.

- **Incident Response:**

- Develop and implement an incident response plan to address security breaches.
- Regularly update the system and apply security patches to mitigate vulnerabilities.

Backup and Recovery Procedures

Backup and recovery procedures are essential to safeguard data against loss due to hardware failures, natural disasters, or cyber-attacks. CGI Advantage 4 provides tools and best practices for creating and managing backups.

- **Backup Strategy:**

- Determine the backup frequency based on the organization's data retention policy.
- Use incremental and full backups to optimize storage and recovery times.

- **Data Backup Process:**

- Schedule automated backups to ensure regular data protection.
- Store backups securely, both on-site and off-site, to prevent data loss.

- **Recovery Procedures:**

- Test backup files regularly to ensure they can be restored successfully.
- Develop a disaster recovery plan outlining steps to restore operations after data loss.

Performance Monitoring

Monitoring System Performance

Monitoring system performance is crucial for ensuring the smooth operation of CGI Advantage 4. This topic covers the tools and techniques for tracking system health and performance metrics.

- **Key Performance Metrics:**
 - Track CPU and memory usage to identify resource bottlenecks.
 - Monitor database performance, including query response times and transaction throughput.
- **Performance Monitoring Tools:**
 - Use built-in monitoring tools to view real-time performance data.
 - Set up alerts for critical performance thresholds, such as high CPU usage or slow query execution.
- **Performance Analysis:**
 - Analyze historical performance data to identify trends and potential issues.
 - Use performance data to inform capacity planning and system scaling decisions.

Troubleshooting Common Issues

Troubleshooting common issues quickly and effectively is essential to maintaining system uptime and user satisfaction. This topic covers common issues that may arise in CGI Advantage 4 and provides strategies for resolving them.

- **Common System Issues:**

- Slow system performance due to high resource usage.
- Database connectivity issues causing application downtime.
- Errors in data processing or report generation.

- **Troubleshooting Strategies:**

- Use system logs to identify the root cause of issues.
- Apply patches or updates to fix known bugs or vulnerabilities.
- Collaborate with vendor support for complex issues that require expert assistance.

- **Preventative Maintenance:**

- Perform regular system audits to identify potential issues before they escalate.
- Update system documentation to reflect changes in configuration or usage patterns.

Optimizing System Performance

Optimizing system performance is an ongoing task that involves fine-tuning various components of CGI Advantage 4 to ensure it operates at peak efficiency. This topic provides best practices for optimizing system performance.

- **System Optimization Techniques:**

- Optimize database queries to reduce load times and improve response rates.
- Allocate system resources based on workload requirements, ensuring that critical processes have sufficient CPU and memory.

- **Load Balancing:**

- Implement load balancing to distribute traffic evenly across servers.
- Use redundancy to ensure high availability and minimize downtime.

- **System Upgrades:**

- Plan and execute system upgrades to take advantage of new features and performance improvements.
- Test upgrades in a staging environment before applying them to the production system.

Module 25: System Setup and Configuration



Learning Objectives



Upon completion of this module, students should be able to:

- Learn how to prepare for the installation of CGI Advantage 4 by understanding system requirements.
- Gain the ability to install CGI Advantage 4 and perform initial configurations.
- Understand how to configure system settings, modules, and user roles to align with organizational needs.

Initial Setup and Installation

System Requirements and Pre-installation Checklist

Before installing CGI Advantage 4, it is crucial to ensure that your environment meets the system requirements. This includes both hardware and software prerequisites, which are essential for the smooth operation of the software.

- **Hardware Requirements:**
 - Processor: Minimum of 4-core CPU (8-core recommended)
 - Memory: At least 16 GB RAM (32 GB recommended)
 - Storage: SSD with at least 500 GB of free space
- **Software Requirements:**
 - Operating System: Windows Server 2016 or higher, or Linux (Red Hat/CentOS preferred)
 - Database: Oracle 19c, SQL Server 2019, or PostgreSQL 12+
 - Web Server: Apache Tomcat 9.x or higher
- **Pre-installation Checklist:**
 - Verify hardware and software requirements
 - Ensure network configurations are in place
 - Backup existing data and systems
 - Prepare a rollback plan in case of issues during installation

Installation Procedures

The installation of CGI Advantage 4 involves several steps, from preparing the environment to executing the installation scripts. Follow the detailed procedures below to ensure a successful installation.

- **Step-by-Step Installation Guide:**

1. Prepare the environment: Install necessary software (e.g., database, web server).
2. Extract the CGI Advantage 4 installation package.
3. Run the installation wizard or execute installation scripts.
4. Configure the database connection and perform initial data load.
5. Set up the web server and deploy the application.
6. Verify the installation by accessing the system and checking logs.

Post-Installation Configuration

After the initial installation, several configuration steps are necessary to tailor the system to your organization's needs. This involves configuring modules, setting up user roles, and fine-tuning system settings.

- **Post-Installation Tasks:**

- Configure system settings: Time zones, language, currency, etc.
- Set up user roles and permissions.
- Configure email notifications and alerts.
- Perform a system health check to ensure all components are functioning properly.

System Configuration

Basic Configuration Settings

Basic configuration settings are critical to ensuring the system operates correctly and efficiently. These settings include general system preferences, user interface options, and core module configurations.

- **Configuring Basic Settings:**
 - System Preferences: Configure general settings like date format, number formats, and currency.
 - User Interface: Customize the UI to meet user needs (themes, layouts, etc.).
 - Module Configuration: Adjust settings for core modules like Financial Management, HR, and Procurement.

Configuring Modules and Features

Each module within CGI Advantage 4 offers a range of configurable features that can be tailored to meet specific business requirements. This section covers the configuration of key modules.

- **Financial Management Configuration:**

- Set up General Ledger accounts and financial periods.
- Configure budget controls and financial reporting structures.

- **Procurement Management Configuration:**

- Define procurement policies and approval workflows.
- Set up vendor profiles and contract management rules.

- **HR and Payroll Configuration:**

- Configure employee classifications, pay schedules, and benefits.
- Set up payroll processing rules and compliance requirements.

User Role and Permission Management

Managing user roles and permissions is crucial to maintaining system security and ensuring that users have the appropriate access levels. CGI Advantage 4 provides a robust role-based access control (RBAC) system.

- **User Role Management:**

- Define roles based on job functions (e.g., Finance Manager, HR Specialist).
- Assign permissions to roles based on required access to modules and features.

- **Permission Management:**

- Create custom permission sets for specific tasks.
- Regularly review and audit user permissions to ensure compliance.

Module : Practical Applications and Case Studies



Learning Objectives



Upon completion of this module, students should be able to:

- Apply the knowledge and skills gained in real-world scenarios through hands-on exercises and case studies.
- Understand how CGI Advantage 4 is implemented in different sectors, including financial management, procurement, and HR automation.
- Learn how to design, optimize, and monitor workflows through practical exercises.

Real-World Case Studies

Case Study 1: Financial Management Implementation

In this case study, we explore the implementation of CGI Advantage 4's Financial Management module in a state government agency. The focus is on how the agency streamlined its financial operations, improved compliance, and enhanced reporting accuracy.

- **Background:**

- Overview of the agency's financial management challenges before implementing CGI Advantage 4.
- Key objectives for the implementation, such as improving budget management and financial reporting.

- **Implementation Process:**

- Steps taken to configure the General Ledger, Accounts Payable, and Budgeting modules.
- Integration with existing financial systems and data migration strategies.

- **Outcomes:**

- Improved accuracy in financial reporting and compliance with state regulations.
- Enhanced budget tracking and forecasting capabilities.
- Reduction in manual processes, leading to time and cost savings.

- **Lessons Learned:**

- Importance of thorough planning and stakeholder engagement in the implementation process.
- Challenges faced during data migration and how they were overcome.

Case Study 2: Procurement and Vendor Management

This case study examines how a large municipal government used CGI Advantage 4 to overhaul its procurement and vendor management processes. The focus is on improving vendor relationships, streamlining procurement workflows, and achieving cost savings.

- **Background:**

- Description of the municipality's procurement challenges, including inefficient vendor management and long procurement cycles.
- Goals for the implementation, such as reducing procurement cycle times and improving vendor performance monitoring.

- **Implementation Process:**

- Steps taken to configure the Procurement and Vendor Management modules.
- Integration with the municipality's financial management system for seamless payment processing.

- **Outcomes:**

- Significant reduction in procurement cycle times and increased efficiency in vendor selection.
- Improved vendor performance tracking and compliance monitoring.
- Cost savings achieved through better contract management and bulk purchasing.

- **Lessons Learned:**

- The role of workflow automation in speeding up procurement processes.
- Importance of training staff on new procurement procedures and tools.

Case Study 3: HR and Payroll Automation

In this case study, we explore how a mid-sized public sector organization implemented CGI Advantage 4 to automate its HR and payroll processes. The focus is on reducing payroll errors, ensuring compliance, and improving employee satisfaction.

- **Background:**

- Overview of the organization's HR and payroll challenges, including frequent payroll errors and compliance issues.
- Objectives for the implementation, such as automating payroll processing and improving HR data management.

- **Implementation Process:**

- Configuration of the HR and Payroll modules, including employee data management and payroll rules.
- Integration with time and attendance systems for accurate payroll calculation.

- **Outcomes:**

- Significant reduction in payroll errors and improved compliance with labor laws.
- Enhanced employee satisfaction due to timely and accurate payroll processing.
- Streamlined HR processes, leading to better management of employee records and benefits.

- **Lessons Learned:**

- Importance of detailed payroll rule configuration to avoid errors.
- Role of integration with other systems in ensuring accurate payroll data.

Hands-On Exercises

Simulated Scenarios

Hands-on exercises in simulated scenarios help learners apply the concepts and techniques learned in previous modules. These exercises cover common tasks in CGI Advantage 4, such as creating reports, managing budgets, and automating workflows.

- **Exercise 1: Creating a Custom Financial Report**
 - Objective: Create a custom financial report using the report designer tool.
 - Instructions: Select data fields, apply filters, and design the report layout.
 - Expected Outcome: A detailed financial report that meets specific departmental needs.
- **Exercise 2: Budget Management Simulation**
 - Objective: Set up and manage a departmental budget, including forecasting and variance analysis.
 - Instructions: Configure budget categories, enter budget amounts, and monitor actual spending against the budget.
 - Expected Outcome: A comprehensive budget management process that tracks financial performance.
- **Exercise 3: Workflow Automation for Procurement**
 - Objective: Design and implement an automated workflow for purchase order approvals.
 - Instructions: Use the workflow designer tool to map out the approval process, define conditions, and assign tasks.
 - Expected Outcome: A fully automated procurement workflow that reduces approval times and improves compliance.

System Configuration Exercises

These exercises focus on configuring various CGI Advantage 4 modules to meet specific organizational needs. Learners will practice setting up modules, managing user roles, and configuring security settings.

- **Exercise 1: Configuring the General Ledger**

- Objective: Set up the General Ledger with appropriate accounts, financial periods, and reporting structures.
- Instructions: Define the chart of accounts, configure financial periods, and set up reporting hierarchies.
- Expected Outcome: A well-organized General Ledger ready for financial transactions and reporting.

- **Exercise 2: User Role and Permission Management**

- Objective: Configure user roles and permissions based on organizational roles.
- Instructions: Create user roles, assign permissions, and configure access controls.
- Expected Outcome: A secure system with user roles that match job responsibilities and access needs.

- **Exercise 3: Security and Compliance Configuration**

- Objective: Set up system security settings to ensure compliance with organizational policies and regulations.
- Instructions: Configure role-based access control, set password policies, and enable multi-factor authentication.
- Expected Outcome: A secure and compliant system configuration that protects sensitive data.

Workflow Creation and Optimization

Workflow creation and optimization exercises help learners practice designing, implementing, and refining workflows in CGI Advantage 4. These exercises emphasize efficiency, accuracy, and compliance.

- **Exercise 1: Designing a Workflow for Financial Approvals**

- Objective: Create a workflow for automating financial approval processes, such as budget approvals or expense authorizations.
- Instructions: Use the workflow designer to map out the approval process, define conditions, and assign roles.
- Expected Outcome: An efficient workflow that automates approvals while maintaining necessary controls.

- **Exercise 2: Optimizing a Procurement Workflow**

- Objective: Review and optimize an existing procurement workflow to improve efficiency and reduce processing times.
- Instructions: Analyze the current workflow, identify bottlenecks, and make adjustments to streamline the process.
- Expected Outcome: A more efficient procurement workflow that speeds up processing and improves vendor relationships.

- **Exercise 3: Monitoring and Refining Workflows**

- Objective: Monitor the performance of an implemented workflow and make refinements based on data and user feedback.
- Instructions: Use workflow analytics to track performance metrics, identify issues, and implement improvements.
- Expected Outcome: A continuously improved workflow that adapts to changing business needs and user requirements.

Module : Best Practices and Future Directions



Learning Objectives



Upon completion of this module, students should be able to:

- Learn best practices for implementing and maintaining CGI Advantage 4 within an organization.
- Understand how to ensure system security and compliance with regulatory standards.
- Explore future trends in ERP technologies and how they apply to CGI Advantage 4, including cloud computing, AI, and blockchain.
- Prepare for future updates and enhancements to stay at the forefront of technology.

Best Practices for CGI Advantage 4

Implementation Best Practices

Successful implementation of CGI Advantage 4 requires careful planning, stakeholder engagement, and adherence to best practices. This topic covers the key steps and strategies to ensure a smooth and effective implementation.

- **Planning and Preparation:**
 - Conduct a thorough needs assessment to understand the organization's requirements.
 - Develop a detailed project plan, including timelines, milestones, and resource allocation.
- **Stakeholder Engagement:**
 - Identify and involve key stakeholders from the beginning to ensure buy-in and support.
 - Establish clear communication channels to keep all parties informed throughout the implementation process.
- **Data Migration and Integration:**
 - Plan and execute data migration carefully to avoid data loss or corruption.
 - Test integrations with existing systems to ensure seamless data exchange.
- **Training and Change Management:**
 - Provide comprehensive training for all users to ensure they are comfortable with the new system.
 - Implement a change management plan to address resistance and ensure a smooth transition.

System Maintenance Best Practices

Maintaining CGI Advantage 4 requires ongoing attention to system performance, security, and user support. This topic outlines best practices for keeping the system running efficiently and securely.

- **Regular Maintenance:**

- Schedule regular maintenance windows to perform updates, patches, and backups.
- Monitor system performance metrics to identify and address potential issues before they escalate.

- **Security and Compliance:**

- Regularly review and update security settings, including user roles, permissions, and authentication methods.
- Conduct periodic security audits to ensure compliance with regulatory requirements.

- **User Support and Training:**

- Provide continuous support and training to users, addressing new features, updates, and system changes.
- Establish a helpdesk or support team to handle user inquiries and technical issues promptly.

- **Documentation and Knowledge Management:**

- Maintain up-to-date documentation on system configurations, workflows, and procedures.
- Create a knowledge base for users to access troubleshooting guides, FAQs, and training materials.

Ensuring System Security and Compliance

Security and compliance are critical components of managing CGI Advantage 4, especially given the sensitive data handled by public sector organizations. This topic covers best practices for ensuring that the system remains secure and compliant with relevant regulations.

- **Security Best Practices:**

- Implement multi-factor authentication (MFA) to enhance system access security.
- Use encryption for data at rest and in transit to protect sensitive information.

- **Compliance Management:**

- Stay informed about relevant regulations and ensure the system complies with them, such as HIPAA, GDPR, or FISMA.
- Conduct regular compliance audits and generate reports to demonstrate adherence to regulatory requirements.

- **Incident Response and Recovery:**

- Develop an incident response plan that outlines procedures for handling security breaches.
- Ensure that backup and recovery procedures are in place and tested regularly to minimize downtime in case of an incident.

Future Trends in ERP Systems

Emerging Trends in ERP Technologies

The ERP landscape is constantly evolving, with new technologies and trends shaping the future of systems like CGI Advantage 4. This topic explores some of the key trends that are likely to impact ERP systems in the coming years.

- **Cloud-Based ERP:**
 - The shift towards cloud-based ERP solutions continues to grow, offering scalability, flexibility, and cost savings.
 - Explore the benefits and challenges of migrating to a cloud-based ERP environment.
- **Artificial Intelligence and Machine Learning:**
 - AI and machine learning are increasingly being integrated into ERP systems to enhance automation, decision-making, and predictive analytics.
 - Discuss the potential applications of AI in financial management, procurement, and HR processes.
- **Mobile and Remote Access:**
 - The demand for mobile and remote access to ERP systems is increasing, driven by the need for flexibility and real-time data access.
 - Explore the features and security considerations of mobile ERP solutions.
- **Blockchain Technology:**
 - Blockchain offers the potential for enhanced transparency, security, and traceability in ERP processes, particularly in supply chain management.
 - Discuss how blockchain could be integrated into CGI Advantage 4 and its potential benefits.

The Future of CGI Advantage

As CGI Advantage continues to evolve, it's important to stay informed about upcoming features, enhancements, and strategic directions. This topic provides insights into the future development roadmap of CGI Advantage 4.

- **Planned Enhancements:**

- Discuss upcoming features and modules that are expected to be introduced in future releases.
- Explore how these enhancements could impact system functionality and user experience.

- **Strategic Direction:**

- Understand CGI's long-term vision for Advantage, including plans for modernization, cloud adoption, and AI integration.
- Consider how these strategic directions align with the organization's goals and objectives.

- **User Community and Feedback:**

- Engage with the CGI Advantage user community to stay informed about best practices, tips, and upcoming changes.
- Provide feedback to CGI to influence future developments and ensure that the system meets evolving organizational needs.

Preparing for Future Updates and Enhancements

Preparing for future updates and enhancements is crucial to ensure that the organization remains at the forefront of technology while minimizing disruption. This topic covers the steps and strategies for managing updates and adopting new features.

- **Update Planning:**

- Develop a plan for testing and deploying updates, including timelines, resources, and communication strategies.
- Identify key stakeholders who need to be involved in the update process, such as IT staff, end users, and management.

- **Testing and Training:**

- Test updates in a staging environment before deploying them to the production system to identify and address potential issues.
- Provide training for users on new features and changes introduced by updates to ensure they can use the system effectively.

- **Change Management:**

- Implement a change management plan to address potential resistance and ensure a smooth transition to new features and enhancements.
- Communicate the benefits of updates and enhancements to users to encourage adoption and engagement.

- **Continuous Improvement:**

- Regularly review the system's performance and user feedback to identify areas for improvement and optimization.
- Stay informed about new developments in ERP technology and consider how they can be leveraged to benefit the organization.

Module 7: Certification and Final Project



Learning Objectives



Upon completion of this module, students should be able to:

- Prepare for CGI Advantage 4 certification by understanding the exam format and key topics.
- Develop a study plan and utilize practice exams to assess readiness for certification.
- Complete a final project that demonstrates mastery of CGI Advantage 4, including configuration, reporting, and workflow automation.
- Present the final project and participate in peer and instructor reviews to refine skills and knowledge.

Preparation for CGI Advantage Certification

Overview of Certification Requirements

Earning certification in CGI Advantage 4 demonstrates proficiency and expertise in using the system effectively. This topic provides an overview of the certification process, including the requirements, exam format, and topics covered.

- **Certification Overview:**

- Description of the certification program, including the levels of certification available (e.g., Associate, Professional).
- Outline of the certification process, including prerequisites, exam registration, and scheduling.

- **Exam Format:**

- Details on the format of the certification exam, including the number of questions, types of questions (e.g., multiple choice, simulation), and time limit.
- Explanation of the scoring system and the passing criteria.

- **Key Exam Topics:**

- Overview of the key topics covered in the exam, such as financial management, procurement, HR, system configuration, and reporting.
- Emphasis on the importance of hands-on experience and practical knowledge in addition to theoretical understanding.

Exam Preparation Tips

Preparing effectively for the CGI Advantage 4 certification exam is crucial for success. This topic offers tips and strategies to help candidates study and prepare for the exam.

- **Study Plan:**

- Develop a personalized study plan that outlines the topics to cover, study materials to use, and timelines for preparation.
- Allocate sufficient time for each topic, focusing more on areas where additional review is needed.

- **Study Resources:**

- Utilize official CGI Advantage 4 documentation, user guides, and training materials to prepare for the exam.
- Engage in hands-on practice by performing tasks in a test environment or using simulation tools.

- **Practice Exams:**

- Take advantage of practice exams to familiarize yourself with the exam format and question types.
- Review incorrect answers to understand areas of weakness and focus on improving those areas.

- **Time Management:**

- Practice managing your time during the exam to ensure you can complete all questions within the allotted time.
- Prioritize answering questions you are confident in first, then return to more challenging ones.

Practice Exams and Sample Questions

Practice exams and sample questions are valuable tools for assessing your readiness for the certification exam. This topic provides access to practice exams and discusses how to use them effectively.

- **Accessing Practice Exams:**

- Information on how to access official practice exams and sample questions provided by CGI or other authorized providers.
- Recommendations for third-party resources that offer additional practice exams and study materials.

- **Using Practice Exams Effectively:**

- Take practice exams under exam-like conditions to simulate the actual testing experience.
- Review your performance on practice exams to identify areas that need further study.

- **Sample Questions:**

- Example questions that cover key exam topics, such as configuring the General Ledger, setting up user roles, and creating custom reports.
- Discussion of the rationale behind each correct answer to reinforce learning.

Final Project

Project Guidelines and Expectations

The final project is an opportunity for learners to demonstrate their mastery of CGI Advantage 4 by applying the skills and knowledge acquired throughout the course. This topic outlines the guidelines, expectations, and evaluation criteria for the final project.

- **Project Objectives:**

- Define the goals of the final project, such as configuring a complete module, creating a comprehensive report, or designing a workflow automation process.
- Emphasize the importance of showcasing both technical skills and problem-solving abilities.

- **Project Scope:**

- Determine the scope of the project, including the specific tasks to be completed and the modules or features to be used.
- Provide flexibility for learners to choose a project that aligns with their interests and professional goals.

- **Submission Requirements:**

- Outline the format for project submission, including documentation, screenshots, and any supporting materials.
- Specify the deadline for project submission and the preferred method for submitting the project (e.g., online portal, email).

Submission and Evaluation Criteria

Understanding the submission process and evaluation criteria is crucial for successfully completing the final project. This topic provides detailed instructions on how to submit the project and what evaluators will be looking for.

- **Submission Process:**

- Step-by-step instructions for submitting the final project, including file formats, required documentation, and any additional information needed.
- Recommendations for ensuring that all components of the project are included and properly formatted.

- **Evaluation Criteria:**

- Explanation of the criteria used to evaluate the final project, including technical accuracy, completeness, creativity, and adherence to project guidelines.
- Breakdown of how the project will be scored, with emphasis on key areas such as configuration accuracy, problem-solving, and documentation quality.

- **Feedback and Revisions:**

- Information on how feedback will be provided to learners after the project is evaluated.
- Guidelines for making revisions if needed and resubmitting the project for final approval.

Final Presentation and Review

The final presentation and review provide an opportunity for learners to showcase their work and receive constructive feedback from instructors and peers. This topic covers the process for presenting the final project and participating in a review session.

- **Presentation Guidelines:**

- Instructions for preparing and delivering a presentation of the final project, including key points to cover and visual aids to use.
- Tips for effectively communicating project objectives, challenges faced, and solutions implemented.

- **Peer Review and Feedback:**

- Overview of the peer review process, where learners can present their projects to classmates and receive feedback.
- Guidelines for providing constructive and supportive feedback to peers.

- **Instructor Review:**

- Explanation of the instructor's role in reviewing the final project and providing detailed feedback.
- Information on how the final project grade will be determined, based on the presentation, project quality, and adherence to guidelines.

- **Celebration of Achievement:**

- Recognition of learners who have successfully completed the course and achieved certification.
- Discussion of next steps for learners, including career opportunities and further learning paths in CGI Advantage 4 and related technologies.

Module 5: Assorted Topics



1. Accounts Receivable

- **Module:** Accounts Receivable
- **Purpose:** Used to record amounts owed by customers, creating a receivable for services rendered or goods provided.
- **Usage:** Primarily used by Accounts Receivable departments to manage and track outstanding amounts due from customers.

Prerequisites and Dependencies

Table	Description
Customer Profile (CPC)	Stores customer information required for billing.
Billing Profile Table	Specifies customer billing terms and preferences.
Revenue Account (RA)	Defines GL accounts used to record receivable amounts.

Key Fields and Data Elements

Field Name	Required/Optional
Description	Invoice Number
Required	Unique identifier for each customer invoice.
Customer ID	Required
Links the transaction to a customer profile.	Account Code
Required	GL receivable account used to track outstanding amounts.
Transaction Date	Required
The date the receivable is recorded.	Amount
Required	Total amount due from the customer.
Payment Terms	Optional

Step-by-Step Process for Creating and Processing RE (Receivable Entry)

1. **Navigate to Receivable Entry:** Go to **Accounts Receivable > Receivable Entry (RE)**.
2. **Enter Header Information:** Include Invoice Number, Date, and Amount.

3. **Link to Customer:** Enter the Customer ID associated with the receivable.
4. **Enter Line Items:** Specify account codes and amounts for each receivable line item.
5. **Save and Validate:** Click **Save** to validate entries. Resolve any validation errors that arise.
6. **Submit for Approval:** If approval is required, submit the transaction through the workflow.
7. **Post Transaction:** Once approved, finalize and post to update the receivable records.

Validation Checks and Error Messages

- **Customer Profile Missing:** Ensure that the Customer ID is active in the Customer Profile Table.
- **Account Code Validation:** Confirm that account codes are set up correctly in the Chart of Accounts.
- **Billing Profile Missing:** Verify that the billing profile is complete for third-party billing scenarios.

Reporting and Monitoring Requirements

- **Transaction Inquiry (TI):** Use this to track and monitor the status of the receivable.
- **Aging Report:** Monitors outstanding receivables and due dates.
- **Approval History:** Check the approval workflow for transaction history and status.

Troubleshooting Common Issues

- **Customer Profile Missing:** Ensure customer records are complete in the VCC table.
- **Account Code Validation Errors:** Ensure valid account codes in COA.
- **Approval Delays:** Confirm that approvers are available for review.

2. AP Invoice (Accounts Payable Invoice)

- **Module:** Accounts Payable
- **Purpose:** Records vendor invoices for expenses.
- **Usage:** Used by Accounts Payable for tracking vendor payments.

Prerequisites and Dependencies

Table	Description
Vendor Profile (VEN)	Stores vendor information.
Budget Control Table	Checks budget availability for expenses.

Table	Description
Chart of Accounts (COA)	Defines GL accounts for AP transactions.

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3. AP Payment

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4. AR Invoice (Accounts Receivable Invoice)

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5. PO (Purchase Order)

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6. PO Amendment

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7. Contract

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8. GL Journal (General Ledger Journal)

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9. Budget Entry (BJE)

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10. Budget Adjustment

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11. Payroll Run (PRR)

...

12. Payroll Adjustment

...

13. Asset Management Entry

...

14. Asset Depreciation

...

15. Vendor Profile Creation (VCC)

...

16. Customer Profile Creation (CPC)

...

17. Billing Profile Setup

...

18. Cash Receipt (CR)

...

19. Time Entry (TME)

...

20. Leave Request (LVR)

...

21. Benefits Enrollment (BEN)

...

22. Budget Control Inquiry (BCI)

...

23. Procurement Bid (BID)

...

24. Revenue Recognition Entry (RRE)

...

25. Receivable Adjustment Entry (RAE)

- **Module:** Accounts Receivable
- **Purpose:** Adjusts previously recorded receivables for corrections or updates.
- **Usage:** Used by AR departments to correct or adjust outstanding receivables.

Prerequisites and Dependencies

Table	Description
Customer Profile (CPC)	Ensures accurate customer information.
Receivable Entry (RE)	Links to existing receivables for adjustment.
Chart of Accounts (COA)	Verifies valid account codes.

Key Fields and Data Elements

Field Name	Required/Optional
Description	Adjustment ID
Required	Unique identifier for the adjustment.
Original RE ID	Required
References the original Receivable Entry.	Adjustment Date
Required	The date the adjustment is recorded.
Amount	Required

Step-by-Step Process for Creating and Processing Receivable Adjustment Entry

1. **Navigate to Receivable Adjustment Entry:** Go to **Accounts Receivable > Adjustment Entry**.

2. **Select Original Receivable:** Find and select the RE to be adjusted.
3. **Enter Adjustment Details:** Provide Adjustment ID, Date, and Amount.
4. **Save and Validate:** Click **Save** to check entries.
5. **Submit for Approval:** If required, submit for approval.
6. **Post Adjustment:** Once approved, finalize and post to update receivable records.

Validation Checks and Error Messages

- **Invalid Adjustment Amount:** Ensure that adjustment does not exceed original amount.
- **Missing Original Receivable:** Confirm the original RE ID is accurate and valid.

Reporting and Monitoring Requirements

- **Adjustment History Report:** Tracks adjustments made to receivables.
- **Transaction Inquiry (TI):** Monitor the adjustment status.

Troubleshooting Common Issues

- **Original RE Missing:** Verify that the referenced RE exists and is active.
- **Approval Delays:** Confirm that assigned approvers are available.

Requirements and Dependencies for Creating and Processing an RE Receivable Transaction Using the Billing Rate Table in CGI Advantage 4

Overview

To create and process an **RE (Receivable Entry) transaction** that uses the **Billing Rate Table** in CGI Advantage 4, there are several system requirements and dependencies. These dependencies ensure that billing rates are applied accurately based on predefined tables, customer profiles, and account structures. This document outlines the requirements, prerequisites, and setup dependencies necessary for successful RE transaction processing.

Requirements and Dependencies

Requirement/Dependency	Description
Customer Profile Setup	A customer profile must be created in the Vendor Customer Creation (VCC) module. This profile includes essential customer information such as name, address, Tax ID, and any billing relationships or payment terms specific to the customer.
Billing Profile Table	The Billing Profile Table must contain a record for each customer requiring third-party billing or special rate applications. The profile should specify the rate schedule and, if applicable, the address ID associated with each customer's billing details.
Billing Rate Table	The Billing Rate Table must be configured with rate codes, rate types, and applicable rate amounts. This table determines the billing rate calculations applied to the RE transaction based on service, customer, or other criteria.
Chart of Accounts (COA)	The Chart of Accounts must be set up with the necessary account codes that correspond to the revenue accounts being billed. This ensures that the receivable entries are allocated correctly within the general ledger.
Revenue Codes	Revenue codes must be created and linked to specific accounts in the Revenue Module . Each revenue code represents a distinct income source and determines how receivable transactions are recorded in the general ledger.
Budget Control and Structure	The budget structure and control settings should be defined for accounts impacted by RE transactions. This may involve setting up budget controls and ensuring that funds are available for the services billed.
Service or Product Codes	If billing is based on specific services or products, these codes must be established and linked to rates in the Billing Rate Table. Each service code should map to a rate and, if applicable, an account code for tracking revenue accurately.

Requirement/Dependency	Description
Customer Account Options (CACT Table)	The Customer Account Options Table (CACT) must define specific billing options for each customer. If Address ID inference is required, the CACT table should include entries that link customers to particular billing addresses and preferences.
User Access and Permissions	Users must have appropriate access rights to create and process RE transactions, view the Billing Rate Table, and access the Billing Profile Table. Role-based permissions should be configured in User Management .
Fiscal Period Setup	The fiscal periods must be defined for the current fiscal year. This ensures that all transactions are recorded in the correct accounting period and that revenue can be tracked accurately over time.
Approval Workflow Configuration	If approvals are required for RE transactions, the approval workflow must be configured to route RE transactions to designated approvers. The workflow should be established in Workflow and Approvals settings.
Tax Codes (if applicable)	If taxes are applied to the receivable transactions, tax codes must be defined. Tax codes should align with the relevant billing rate and service/product codes.
Billing Frequency and Payment Terms	Billing frequency (e.g., monthly, quarterly) and payment terms should be set up within the Billing Profile associated with each customer. This helps determine the timing and terms under which the customer will be billed and expected to pay.
System Configuration Settings	System settings must allow for the use of the Billing Rate Table in RE transactions. This may include enabling specific modules or transaction types and ensuring that dependencies like Rate Codes are active.

Process Overview

Once all dependencies are met, the following steps outline the process for creating and processing an RE transaction with the Billing Rate Table:

- Create or Verify Customer Profile:** Ensure the customer profile exists in VCC with relevant billing information and payment terms.
- Set Up Billing Profile:** Define or verify the billing profile in the Billing Profile Table, specifying the Billing Rate Table reference.
- Create RE Transaction:**
 - Go to **Revenue Entry (RE)** and start a new transaction.
 - Select the **Customer ID** and apply the **Billing Rate Table** for rate calculations.
 - Enter required line items, specifying revenue codes, service/product codes, and rate codes as needed.
- Verify Budget and Funds:** Use the Budget Inquiry to confirm available budget for accounts affected by the transaction.
- Submit for Approval:** If approvals are configured, submit the RE transaction for routing through the established workflow.

6. **Monitor and Complete Transaction:** Upon approval, the RE transaction is processed and recorded. Monitor transaction status to ensure completion.

Notes

- **Dependencies Verification:** Confirm all tables, profiles, and rate codes are updated and accurate before initiating an RE transaction.
- **Role Permissions:** Ensure users have the necessary access to the Billing Rate Table and related RE transaction components.
- **Rate Calculation:** Billing Rate Table settings determine the calculation method for applying rates in RE transactions. Verify that rate configurations align with organizational billing practices.

Here's the documentation for creating an **RE (Receivable Entry) transaction** for a customer with multiple third-party billing options in CGI Advantage 4, formatted as an **.adoc** file.

Requirements and Dependencies for Creating an RE Transaction for a Customer with Multiple Third-Party Billing Options in CGI Advantage 4

Overview

In CGI Advantage 4, you can create an **RE (Receivable Entry) transaction** for a customer who requires multiple third-party billing options. This allows a primary customer to receive services while designated third-party entities are billed on their behalf. This setup involves specific configuration in customer profiles, billing profiles, and address tables. This document outlines the requirements, dependencies, and configuration steps to process RE transactions under this arrangement.

Requirements and Dependencies

Requirement/Dependency	Description
Customer Profile Setup	Create or verify a profile for the primary customer in Vendor Customer Creation (VCC) . This profile represents the customer receiving the service and links to the billing profiles and options for multiple third-party payers.
Billing Profile Table	Define multiple billing profiles in the Billing Profile Table for the customer. Each billing profile specifies a unique third-party payer and links to a corresponding Address ID for the designated billing recipient.
Third-Party Customer Profiles	Create profiles for each third-party customer responsible for paying bills on behalf of the primary customer. Each third-party customer should have a record in VCC , including address and contact information.
Billing Rate Table (if applicable)	If billing is based on rates, configure the Billing Rate Table with rate codes and applicable rates. These rates should align with the service or product codes and may vary by third-party customer as needed.
Chart of Accounts (COA)	Set up the Chart of Accounts to define revenue accounts impacted by the RE transaction. This ensures the receivables are accurately recorded in the general ledger.
Revenue Codes	Create and assign revenue codes to specific accounts within the Revenue Module . Revenue codes represent distinct income sources and will be referenced in the RE transaction.
Service or Product Codes	If billing varies by service or product, these codes must be established in the system and linked to the Billing Rate Table (if applicable). Each service or product code should map to an account code.

Requirement/Dependency	Description
Customer Account Options Table (CACT)	Use the Customer Account Options Table (CACT) to define options for multiple billing addresses and payers. Each third-party payer must have an entry in the CACT table that links the primary customer to specific billing preferences and addresses.
User Access and Permissions	Ensure users have appropriate permissions to create and process RE transactions, access the Billing Profile Table, and set up CACT configurations for multiple billing options. Permissions should be managed through User Management .
Fiscal Period Setup	Define fiscal periods for the current fiscal year to ensure transactions are recorded in the correct accounting period.
Approval Workflow Configuration	Configure approval workflows, if required, for RE transactions involving third-party billing. Approval routing should be set up in the Workflow and Approvals section.
Tax Codes (if applicable)	If taxes apply to the transaction, tax codes must be created in alignment with relevant service or product codes.
Payment Terms and Billing Frequency	Define payment terms and billing frequencies within each billing profile for third-party customers. This setup determines the terms under which third parties will be billed and expected to pay.
System Configuration Settings	Confirm that system settings enable third-party billing for RE transactions. Ensure that dependencies such as CACT and Billing Profile configurations are activated.

Process Overview

Once all dependencies are met, follow these steps to create and process an RE transaction with multiple third-party billing options:

- Create or Verify Customer Profile:** Ensure the primary customer's profile exists in VCC with relevant billing information, and confirm each third-party payer profile is complete with contact and billing address details.
- Define Multiple Billing Profiles:** In the **Billing Profile Table**, create separate billing profiles for each third-party customer linked to the primary customer. Specify unique billing addresses using the Address ID linked to each third party in the CACT table.
- Link Third-Party Payers via CACT Table:** Set up entries in the **Customer Account Options Table (CACT)** for each third-party payer. This configuration allows the system to infer the correct Address ID and billing preferences when creating the RE transaction.
- Create RE Transaction:**
 - Navigate to **Revenue Entry (RE)** and initiate a new transaction.
 - Select the primary customer's **Customer ID**.
 - Apply the appropriate billing profile for each line item that reflects the designated third-party payer.
 - Specify line item details such as revenue codes, service or product codes, and, if applicable, rate codes from the Billing Rate Table.

5. **Verify Budget and Available Funds:** Use Budget Inquiry to confirm available budget for accounts involved in the transaction.
6. **Submit for Approval:** If approval workflows are configured, submit the RE transaction to route it through the designated approvers.
7. **Monitor and Complete Transaction:** Upon approval, the RE transaction will be processed with each line item billed to the assigned third-party payer. Monitor transaction status for completion and invoice generation.

Notes

- **Billing Profile and CACT Linkage:** Ensure that each billing profile is linked accurately to the third-party customer's Address ID in the CACT table to avoid billing errors.
- **Role Permissions:** Verify that users have the necessary permissions to set up and view third-party billing profiles and access customer account options.
- **Rate Calculations:** When using the Billing Rate Table, confirm that rate configurations align with the service or product being billed and are appropriate for each third-party payer.

This configuration allows for flexible billing to multiple third-party customers on behalf of a primary customer, streamlining the receivables process in CGI Advantage 4.

Module 8: Core Functionalities



Learning Objectives



Upon completion of this module, students should be able to:

- Master the setup and management of the General Ledger, Accounts Payable, and Accounts Receivable modules.
- Learn how to create and manage budgets and forecasts within CGI Advantage 4.
- Gain proficiency in managing procurement processes, including vendor management and contract creation.
- Understand the functionalities of the HR and Payroll modules, including employee data management and payroll processing.

Financial Management

General Ledger Setup and Management

The General Ledger (GL) is the central repository for financial data in CGI Advantage 4. It tracks all financial transactions and provides the foundation for financial reporting. Proper setup and management of the GL are essential for accurate financial records.

- **Setting Up the General Ledger:**
 - Chart of Accounts: Define the structure of accounts and categories.
 - Financial Periods: Set up fiscal year, periods, and accounting calendar.
 - Journal Entries: Configure rules for automated and manual journal entries.
- **Managing the General Ledger:**
 - Posting Transactions: Ensure timely and accurate posting of transactions.
 - Reconciliation: Perform regular reconciliations to maintain accuracy.
 - Financial Reporting: Generate trial balances, balance sheets, and income statements.

Accounts Payable and Receivable

Accounts Payable (AP) and Accounts Receivable (AR) are critical components of financial management. AP manages outgoing payments to vendors, while AR tracks incoming payments from customers or other entities.

- **Accounts Payable:**

- Vendor Setup: Configure vendor profiles and payment terms.
- Invoice Processing: Automate and manage invoice approvals and payments.
- Payment Runs: Schedule and execute payment batches.

- **Accounts Receivable:**

- Customer Setup: Define customer profiles and payment terms.
- Invoice Generation: Automate the creation and distribution of invoices.
- Collections Management: Track overdue payments and manage collections.

Budgeting and Forecasting

Budgeting and forecasting are vital for financial planning and control. CGI Advantage 4 provides tools to create, manage, and analyze budgets and forecasts, helping organizations to allocate resources efficiently.

- **Budget Setup:**

- Budget Structures: Define budget categories and hierarchies.
- Budget Entry: Input budget amounts for various departments and projects.

- **Forecasting:**

- Revenue and Expense Forecasting: Use historical data to predict future financial performance.
- Scenario Analysis: Create multiple budget scenarios for different assumptions.

- **Monitoring and Reporting:**

- Budget vs. Actual: Track actual performance against the budget.
- Variance Analysis: Identify and analyze discrepancies between budgeted and actual figures.

Procurement Management

Procurement Process Overview

Procurement management in CGI Advantage 4 involves acquiring goods and services to meet the organization's needs. This process includes requisitioning, vendor selection, contract management, and purchase order creation.

- **Procurement Lifecycle:**
 - Requisitioning: Request goods and services through requisition forms.
 - Vendor Selection: Evaluate and select vendors based on predefined criteria.
 - Contract Management: Manage contracts, including terms, renewals, and compliance.
 - Purchase Orders: Generate and manage purchase orders (POs) for approved requisitions.

Vendor Management

Effective vendor management ensures that the organization works with reliable vendors, gets the best value, and maintains strong supplier relationships.

- **Vendor Registration and Approval:**

- Register new vendors in the system.
- Perform vendor due diligence and approval processes.

- **Vendor Performance Monitoring:**

- Track vendor performance metrics, including delivery times and quality.
- Manage vendor ratings and feedback for continuous improvement.

- **Vendor Compliance:**

- Ensure vendors meet compliance requirements, such as certifications and insurance.
- Manage documentation and audits for vendor compliance.

Contract and Purchase Order Management

Managing contracts and purchase orders effectively is crucial for controlling procurement costs and ensuring compliance with organizational policies.

- **Contract Management:**

- Contract Creation: Draft and approve contracts using standardized templates.
- Contract Monitoring: Track contract performance, renewals, and amendments.

- **Purchase Order Management:**

- PO Creation: Automate the creation of POs from approved requisitions.
- PO Tracking: Monitor the status of POs from issuance to fulfillment.
- Invoice Matching: Match POs with invoices for accurate payment processing.

Human Resources and Payroll

HR Module Overview

The Human Resources (HR) module in CGI Advantage 4 is designed to manage employee data, streamline HR processes, and ensure compliance with employment regulations. It covers employee lifecycle management, benefits administration, and compliance reporting.

- **HR Module Components:**

- **Employee Data Management:** Centralized repository for employee information.
- **Benefits Administration:** Manage employee benefits, including health, retirement, and other perks.
- **Compliance Reporting:** Generate reports to ensure compliance with labor laws and regulations.

Employee Data Management

Effective employee data management is essential for accurate HR operations and compliance. This topic covers the setup and maintenance of employee records, including personal information, employment history, and job classifications.

- **Employee Record Setup:**

- Create and update employee profiles with personal and job-related information.
- Manage employment status, job classifications, and department assignments.

- **Employee Lifecycle Management:**

- Onboarding: Automate the onboarding process, including document collection and training.
- Transfers and Promotions: Manage employee transfers, promotions, and job reclassifications.
- Offboarding: Handle employee exits, including exit interviews and final pay processing.

Payroll Processing and Compliance

Payroll processing involves calculating employee wages, withholding taxes, and ensuring compliance with labor laws. CGI Advantage 4 automates these processes, reducing errors and ensuring timely payments.

- **Payroll Setup:**

- Pay Schedules: Define pay periods and pay dates for different employee groups.
- Payroll Elements: Configure earnings, deductions, and benefits components.

- **Payroll Processing:**

- Calculate gross and net pay, including overtime, bonuses, and deductions.
- Generate payroll reports for auditing and compliance purposes.

- **Compliance and Reporting:**

- Ensure payroll processes comply with local, state, and federal regulations.
- Generate required tax forms, such as W-2s and 1099s, for employees and contractors.

Module : Extensibility in CGI Advantage 4



Module Overview

This module explores the concept of extensibility in CGI Advantage 4, providing insights into how the platform can be customized and extended to meet specific organizational requirements. Key topics include the extensibility framework, customization capabilities, and best practices.

Learning Objectives

- Understand the extensibility framework in CGI Advantage 4.
- Identify key areas where customization and extension are possible.
- Apply best practices for maintaining and upgrading customizations.
- Explore real-world examples of extensibility in action.

Lesson 1: Introduction to Extensibility in CGI Advantage 4

Lesson Summary

Extensibility in CGI Advantage 4 allows organizations to tailor the software to their unique business needs without compromising the core functionality. This lesson provides an overview of the extensibility framework and its importance.

Key Topics

- Overview of the extensibility framework.
- Benefits of extensibility.

- Introduction to extensibility tools and techniques.

Detailed Content

CGI Advantage 4 provides a robust extensibility framework that includes:

- **Configuration Options:** Modify system settings, workflows, and processes through user-friendly interfaces.
- **Extension Points:** Add new functionality using APIs, event hooks, and modular designs.
- **Custom Development:** Develop bespoke solutions using supported programming languages and tools.
- **Integration Frameworks:** Connect CGI Advantage with third-party systems via standard protocols like REST and SOAP.

Exercises

1. List three benefits of leveraging extensibility in CGI Advantage 4.
2. Identify one business process in your organization that could benefit from customization.

Lesson 2: Areas of Extensibility

Lesson Summary

This lesson delves into the specific areas within CGI Advantage 4 that support customization and extensibility.

Key Topics

- Workflow customization.
- Screen and UI extensions.
- Business rule configuration.
- Integration with external systems.

Detailed Content

Workflow Customization

- Use the Workflow Designer to modify or create new approval workflows.
- Add custom logic to workflows using predefined triggers.

Screen and UI Extensions

- Modify existing screens or create new ones using the UI Designer.
- Add custom fields, dropdowns, or tabs to enhance user experience.

Business Rule Configuration

- Use the Rules Engine to define or override default business rules.
- Create custom validation or calculation rules.

Integration with External Systems

- Utilize Integration Manager to set up and manage external system connections.
- Leverage APIs and web services for data exchange.

Exercises

1. Design a simple workflow for a new approval process.
2. Identify which screen in your organization would benefit from additional fields or tabs.

Lesson 3: Tools and Techniques for Extensibility

Lesson Summary

CGI Advantage 4 offers a suite of tools and techniques for implementing extensibility. This lesson covers the tools available and their use cases.

Key Topics

- Configuration tools.
- Development tools.
- Testing and deployment practices.

Detailed Content

Configuration Tools

- Admin Console: Used for configuring system settings, workflows, and user roles.
- UI Designer: Enables screen modifications.

Development Tools

- Extension Manager: Manage and deploy extensions.
- Developer Toolkit: Provides APIs, SDKs, and sample code.

Testing and Deployment Practices

- Test customizations in a sandbox environment.
- Use version control systems to manage changes.
- Deploy extensions incrementally to minimize disruptions.

Exercises

1. Use the Admin Console to create a test configuration for a new role.
2. Explore the Developer Toolkit to create a sample extension.

Lesson 4: Best Practices for Extensibility

Lesson Summary

This lesson highlights best practices to ensure maintainable and upgrade-friendly extensibility.

Key Topics

- Planning extensibility projects.
- Documentation and version control.
- Maintaining upgrade compatibility.

Detailed Content

Planning Extensibility Projects

- Define clear objectives for customization.
- Identify stakeholders and involve them in the design process.

Documentation and Version Control

- Maintain detailed documentation for all customizations.
- Use version control systems to track changes.

Maintaining Upgrade Compatibility

- Follow CGI's guidelines for customizations.
- Avoid modifying core code to ensure compatibility with future updates.

Exercises

1. Create a checklist for planning an extensibility project.
2. Document a sample customization using version control.

Lesson 5: Case Studies and Real-World Applications

Lesson Summary

This lesson showcases how extensibility is used in real-world scenarios to address unique business challenges.

Key Topics

- Case study: Customizing a procurement workflow.
- Case study: Integrating with a financial management system.

Detailed Content

Customizing a Procurement Workflow

- Scenario: An organization requires an additional approval step for high-value purchases.
- Solution: Using Workflow Designer to add a custom approval step.

Integrating with a Financial Management System

- Scenario: Automating data exchange between CGI Advantage and an external financial system.
- Solution: Leveraging Integration Manager to implement a REST API connection.

Exercises

1. Analyze the provided case study and suggest additional enhancements.
2. Develop a simple integration plan for a hypothetical system.

Module Summary

In this module, we explored the extensibility framework in CGI Advantage 4, including key areas, tools, and best practices. By understanding and applying these concepts, organizations can maximize the value of CGI Advantage while ensuring maintainability and upgradeability.

Next Steps

- Practice using the extensibility tools in a sandbox environment.

- Explore additional case studies to deepen your understanding of real-world applications.

Module 20: Reporting and Analytics



Learning Objectives



Upon completion of this module, students should be able to:

- Master the use of reporting tools within CGI Advantage 4 to create, customize, and distribute reports.
- Gain skills in data analytics and business intelligence to drive data-driven decision-making.
- Learn how to implement and utilize data visualization techniques to effectively present business insights.

Reporting Tools and Features

Overview of Reporting Tools

CGI Advantage 4 offers a comprehensive set of reporting tools that allow users to generate, customize, and distribute reports across various modules. These tools support both standard and ad hoc reporting needs, enabling users to extract actionable insights from the system.

- **Key Reporting Tools:**

- Standard Reports: Pre-built reports covering common financial, procurement, and HR scenarios.
- Custom Reports: Tools for designing and building reports tailored to specific business requirements.
- Ad Hoc Reporting: On-the-fly report generation based on user-defined criteria.

- **Report Formats:**

- Generate reports in multiple formats, including PDF, Excel, CSV, and HTML.
- Schedule automated report generation and delivery via email or system notifications.

- **Report Distribution:**

- Share reports securely with stakeholders through the system's distribution features.
- Set up report access controls to ensure only authorized users can view or modify reports.

Creating and Customizing Reports

Creating and customizing reports in CGI Advantage 4 allows organizations to meet their specific reporting needs. This topic covers the steps to design, customize, and refine reports using the system's reporting tools.

- **Report Design:**

- Use the report designer interface to select data sources, fields, and layout options.
- Implement sorting, grouping, and filtering options to organize report data effectively.

- **Customizing Reports:**

- Add custom calculations, formulas, and conditional formatting to enhance report functionality.
- Modify report templates to align with the organization's branding and style guidelines.

- **Report Templates:**

- Save customized reports as templates for reuse across the organization.
- Create a library of report templates for different departments or use cases.

Scheduled and Ad Hoc Reporting

Scheduled and ad hoc reporting features in CGI Advantage 4 allow users to automate regular report generation and quickly produce reports on demand. This topic explains how to configure and manage these reporting options.

- **Scheduled Reporting:**
 - Set up recurring reports to be generated at specific intervals (daily, weekly, monthly).
 - Configure report delivery options, including email, system notifications, and secure file transfer.
- **Ad Hoc Reporting:**
 - Use the ad hoc reporting tool to create custom reports in response to immediate needs.
 - Save ad hoc reports for future reference or modification.
- **Managing Reports:**
 - Monitor scheduled report jobs to ensure timely execution and delivery.
 - Archive old reports to maintain system performance and comply with data retention policies.

Data Analytics and Business Intelligence

Introduction to Data Analytics

Data analytics involves examining data sets to draw conclusions about the information they contain. CGI Advantage 4 integrates data analytics tools that help organizations make data-driven decisions by identifying trends, patterns, and insights within their operational data.

- **Types of Data Analytics:**

- Descriptive Analytics: Summarizes historical data to understand what has happened.
- Predictive Analytics: Uses historical data to predict future outcomes.
- Prescriptive Analytics: Recommends actions based on data-driven insights.

- **Data Sources:**

- Leverage data from various CGI Advantage 4 modules, including financial, HR, and procurement.
- Integrate external data sources to enrich analysis and provide a more comprehensive view.

- **Analytical Tools:**

- Use built-in tools for data visualization, trend analysis, and forecasting.
- Implement advanced analytical techniques using third-party tools integrated with CGI Advantage 4.

Implementing Business Intelligence Solutions

Business Intelligence (BI) solutions in CGI Advantage 4 empower organizations to transform data into actionable insights. This topic covers the implementation of BI tools and techniques within the system.

- **BI Tools Overview:**

- Explore built-in BI tools for creating dashboards, scorecards, and KPIs.
- Integrate with external BI platforms (e.g., Power BI, Tableau) for enhanced analytics.

- **Setting Up Dashboards:**

- Design and configure dashboards that provide real-time insights into key performance indicators (KPIs).
- Customize dashboards to reflect the specific needs of different departments or roles.

- **Data Warehousing:**

- Implement data warehousing strategies to centralize and standardize data for analysis.
- Use ETL (Extract, Transform, Load) processes to prepare data for BI applications.

Data Visualization Techniques

Data visualization is a critical component of data analytics, allowing users to see trends and patterns that might not be immediately apparent in raw data. CGI Advantage 4 provides a variety of data visualization tools and techniques.

- **Choosing the Right Visualization:**

- Select appropriate visualization types (e.g., charts, graphs, heatmaps) based on the data and analysis goals.
- Use dashboards to combine multiple visualizations for a comprehensive view of business metrics.

- **Designing Effective Visualizations:**

- Apply design principles to create clear, informative, and visually appealing charts and graphs.
- Use color, labeling, and interactive elements to enhance the user experience.

- **Sharing Visualizations:**

- Embed visualizations in reports, dashboards, and presentations for easy sharing with stakeholders.
- Export visualizations in various formats for use in external documents or systems.

Module : CGI Advantage 4 Core Transaction Documentation



This document provides comprehensive details for 25 core transactions within CGI Advantage 4. Each transaction includes an overview, prerequisites, dependencies, process steps, validation checks, reporting, and troubleshooting.

1. AP Invoice (Accounts Payable Invoice)

- **Module:** Accounts Payable
- **Purpose:** Used to record vendor invoices for expenses incurred.
- **Usage:** Primarily used by Accounts Payable departments to track vendor invoices.

Prerequisites and Dependencies

Table	Description
Vendor Profile (VEN)	Stores vendor information.
Budget Control Table	Checks budget availability.

Key Fields and Data Elements

Field Name	Required/Optional	Description
Invoice Number	Required	Unique identifier for each invoice.
Vendor ID	Required	Links the transaction to a vendor profile.
Account Code	Required	GL account used for the expense.

Process for Creating and Processing AP Invoice

1. **Navigate to AP Invoice:** Go to **Accounts Payable > Invoice**.
2. **Enter Header Information:** Include Invoice Number, Date, and Amount.
3. **Link to Vendor:** Enter Vendor ID.
4. **Enter Line Items:** Specify account codes and amounts.
5. **Submit for Approval:** Submit the transaction.

6. **Post Transaction:** Finalize and post.

Validation and Troubleshooting

- **Budget Errors:** Check the Budget Control Table for availability.
- **Vendor Profile Missing:** Ensure Vendor ID is set up.

2. AP Payment

- **Module:** Accounts Payable
- **Purpose:** Processes payments for recorded invoices.
- **Usage:** Used by AP departments to issue payments to vendors.

...

3. AR Invoice (Accounts Receivable Invoice)

- **Module:** Accounts Receivable
- **Purpose:** Records receivables and customer invoices.
- **Usage:** AR departments use it to manage billing and incoming payments.

...

4. AR Payment

- **Module:** Accounts Receivable
- **Purpose:** Applies customer payments to outstanding invoices.
- **Usage:** Used by AR departments for cash receipts and payment application.

...

5. RE (Revenue Entry)

- **Module:** Revenue
- **Purpose:** Records revenue transactions.
- **Usage:** Finance departments use it for income tracking.

...

6. PO (Purchase Order)

- **Module:** Procurement
- **Purpose:** Creates purchase orders for goods and services.
- **Usage:** Procurement teams use it to initiate purchase agreements.

...

7. PO Amendment

- **Module:** Procurement
- **Purpose:** Modifies existing purchase orders.
- **Usage:** Used when changes to quantity or terms are needed.

...

8. Contract

- **Module:** Procurement
- **Purpose:** Manages contracts with vendors.
- **Usage:** Used by procurement and legal for vendor agreements.

...

9. GL Journal (General Ledger Journal)

- **Module:** General Ledger
- **Purpose:** Records adjustments and journal entries in GL.
- **Usage:** Finance teams use it for financial reconciliation.

...

10. Budget Entry (BJE)

- **Module:** Budget
- **Purpose:** Enters budget allocations.
- **Usage:** Budgeting departments use it for planning and control.

...

11. Budget Adjustment

- **Module:** Budget
- **Purpose:** Modifies existing budget entries.
- **Usage:** Used for budget reallocations or adjustments.

...

12. Payroll Run (PRR)

- **Module:** Payroll
- **Purpose:** Processes payroll for employees.
- **Usage:** Payroll departments use it to manage salaries.

...

13. Payroll Adjustment

- **Module:** Payroll
- **Purpose:** Adjusts payroll entries.
- **Usage:** Used for corrections or re-calculations.

...

14. Asset Management Entry

- **Module:** Asset Management
- **Purpose:** Records assets in the system.
- **Usage:** Finance uses it to track physical assets.

...

15. Asset Depreciation

- **Module:** Asset Management
- **Purpose:** Calculates depreciation for assets.

- **Usage:** Used in financial reporting.

...

16. Vendor Profile Creation (VCC)

- **Module:** Vendor/Customer
- **Purpose:** Adds vendor information to the system.
- **Usage:** AP and Procurement departments use it for vendor records.

...

17. Customer Profile Creation (CPC)

- **Module:** Vendor/Customer
- **Purpose:** Adds customer information.
- **Usage:** AR and Revenue departments use it for customer records.

...

18. Billing Profile Setup

- **Module:** Vendor/Customer
- **Purpose:** Defines customer billing preferences.
- **Usage:** Linked to AR for customer billing.

...

19. Cash Receipt (CR)

- **Module:** Accounts Receivable
- **Purpose:** Records incoming cash.
- **Usage:** Used to track payments received.

...

20. Time Entry (TME)

- **Module:** Payroll

- **Purpose:** Logs employee time for payroll.
- **Usage:** Used by HR and Payroll for time tracking.

...

21. Leave Request (LVR)

- **Module:** Payroll
- **Purpose:** Manages employee leave requests.
- **Usage:** HR departments use it for leave tracking.

...

22. Benefits Enrollment (BEN)

- **Module:** HR
- **Purpose:** Enrolls employees in benefits plans.
- **Usage:** HR uses it for benefits administration.

...

23. Budget Control Inquiry (BCI)

- **Module:** Budget
- **Purpose:** Checks budget controls.
- **Usage:** Used by Finance to monitor budget adherence.

...

24. Procurement Bid (BID)

- **Module:** Procurement
- **Purpose:** Manages bids for procurement.
- **Usage:** Procurement uses it to evaluate suppliers.

...

25. Revenue Recognition Entry (RRE)

- **Module:** Revenue
- **Purpose:** Records revenue recognition.
- **Usage:** Used in compliance and reporting.

Module : Core Transaction

Documentation for CGI Advantage 4



Template Overview

This document provides a template for documenting core transactions in CGI Advantage 4. Each core transaction section includes setup requirements, dependencies, processing steps, validation checks, and any key reporting or monitoring requirements.

Transaction Documentation Template



Replace placeholders (e.g., [\[Transaction Name\]](#)) with specific transaction details for each core transaction.

1. Transaction Overview: [\[Transaction Name\]](#)

- **Module:** [\[Module Name\]](#)
- **Purpose:** Briefly describe the purpose of the transaction, e.g., "The **AP Invoice** transaction is used to record payable invoices within the Accounts Payable module."
- **Usage:** Specify the typical use cases and departments that commonly use this transaction.

2. Prerequisites and Dependencies

Required Configurations

- **Chart of Accounts (COA):** Ensure all relevant GL accounts are established for this transaction type.
- **Customer/Vendor Profiles:** Make sure customers/vendors are correctly set up in the VCC table.
- **Fiscal Year Setup:** Confirm the fiscal period is open for transaction processing.

Related Tables and Codes

Table	Description
[Primary Table Name]	This table holds the primary data for the transaction, such as AP Invoice (API) or Revenue Entry (RE).
[Related Table 1]	Contains related data, such as Vendor Profile or Customer Account.
[Related Table 2]	Stores related settings, such as Billing Profile or Budget Control.

3. Key Fields and Data Elements

This section outlines the most critical fields in the transaction form and their respective requirements.

Field Name	Required/Optional	Description
Transaction ID	Required	Unique identifier for each transaction, usually system-generated.
Transaction Date	Required	The date the transaction is recorded in the system.
Account Code	Required	GL account used for the transaction, must match entries in the COA.
Amount	Required	The total amount of the transaction.
Customer/Vendor ID	Required	Links the transaction to the correct customer/vendor profile.
Approval Workflow	Optional	Indicates if approval is needed before posting.

4. Step-by-Step Process for Creating and Processing [Transaction Name]

1. **Navigate to Transaction:** Go to [Module] > [Submodule] > [Transaction Name].
2. **Enter Transaction Header Information:**

a. Enter the Transaction Date.

b. Provide a Description of the transaction.

c. Enter the Transaction Amount.
3. **Enter Line Items:**

a. Specify each line item, including Account Code, Amount, and any necessary Service Codes.
4. **Link to Customer/Vendor:**

a. Enter the Customer/Vendor ID to associate the transaction with the relevant party.
5. **Save and Validate:** Click Save to validate the entries. Resolve any validation errors that arise.

6. **Submit for Approval** (if applicable): If approval is required, submit the transaction through the workflow.
7. **Post Transaction**: Once approved, post the transaction to update financial records.

5. Key Validation Checks and Error Messages

- **Missing Account Code**: Ensure that all account codes are linked to the COA.
- **Budget Availability**: Confirm sufficient budget exists for expense-related transactions.
- **Customer/Vendor Profile Missing**: Verify that the Customer/Vendor ID is active and complete.
- **Address ID Required**: When using third-party billing, the Address ID must be set in the CACT.

6. Reporting and Monitoring Requirements

- **Transaction Inquiry (TI)**: Use this to track and monitor the transaction status.
- **Approval History**: Check the approval workflow for transaction history and status.
- **Custom Reports**: If needed, custom reports can be configured to show specific transaction data (e.g., Vendor Reports for AP transactions).

7. Related Transactions and Follow-Up Actions

Related Transaction	Description
[Related Transaction 1]	For example, AP Payment following an AP Invoice.
[Related Transaction 2]	For example, AR Payment after an AR Invoice.

8. Troubleshooting Common Issues

- **Budget Errors**: Verify the budget setup and ensure sufficient funds.
- **Approval Delays**: Check workflow configurations and confirm approver assignments.
- **Data Validation Errors**: Ensure all required fields are populated and formatted correctly.

Module : Core Tables in HRM and FIN Modules for CGI Advantage 4



HRM Core Tables

1. Employee Profile (EMP)

Employee ID	First Name	Last Name	Department Code	Job Code	Pay Grade	Status
1001	John	Doe	HR01	HRMGR	P5	Active
1002	Jane	Smith	FIN02	FINACCT	P3	Active
1003	Alice	Johnson	IT03	ITDEV	P4	Inactive
1004	Bob	Lee	HR01	HRASST	P2	Active

2. Employee Address (EAD)

Employee ID	Address Type	Address Line 1	City	State	Zip Code
1001	Home	123 Elm St.	Anytown	CA	90001
1002	Work	456 Oak Ave.	Big City	NY	10010
1003	Home	789 Pine Rd.	Smallville	TX	75001
1004	Work	321 Maple Blvd.	Capital	DC	20001

3. Payroll Run (PRR)

Payroll Run ID	Pay Period Start	Pay Period End	Pay Date	Status
PR202401	2024-01-01	2024-01-15	2024-01-20	Closed
PR202402	2024-01-16	2024-01-31	2024-02-05	Closed
PR202403	2024-02-01	2024-02-15	2024-02-20	Pending
PR202404	2024-02-16	2024-02-29	2024-03-05	Scheduled

4. Benefits Enrollment (ENR)

Employee ID	Benefit Plan ID	Enrollment Date	Coverage Level	Status
1001	MED01	2023-01-01	Family	Active

Employee ID	Benefit Plan ID	Enrollment Date	Coverage Level	Status
1002	DEN01	2023-02-15	Individual	Active
1003	VSN01	2023-03-01	Family	Inactive
1004	MED01	2023-04-01	Individual	Active

5. Time Entry (TME)

Time Entry ID	Employee ID	Date	Hours Worked	Time Code	Approval Status
TE10001	1001	2024-01-10	8	REG	Approved
TE10002	1002	2024-01-11	8	REG	Pending
TE10003	1003	2024-01-12	6	SICK	Approved
TE10004	1004	2024-01-13	4	VAC	Rejected

FIN Core Tables

1. General Ledger Account (GAC)

Account ID	Account Name	Account Type	Fund Code	Department Code
5000	Salaries Expense	Expense	GF01	HR01
6000	Office Supplies	Expense	GF02	FIN02
7000	IT Equipment	Expense	GF03	IT03
8000	Training Budget	Expense	GF01	HR01

2. Accounts Payable Invoice (API)

Invoice ID	Vendor ID	Invoice Date	Due Date	Amount	Status
INV1001	VEND001	2024-01-01	2024-01-15	1000.00	Paid
INV1002	VEND002	2024-01-10	2024-01-25	500.00	Pending
INV1003	VEND003	2024-01-12	2024-01-30	750.00	Approved
INV1004	VEND004	2024-01-15	2024-01-31	200.00	Rejected

3. Accounts Receivable Invoice (ARI)

Invoice ID	Customer ID	Invoice Date	Due Date	Amount	Status
ARINV1001	CUST001	2024-01-05	2024-01-20	1500.00	Open
ARINV1002	CUST002	2024-01-07	2024-01-22	2500.00	Closed
ARINV1003	CUST003	2024-01-10	2024-01-25	1800.00	Open
ARINV1004	CUST004	2024-01-15	2024-01-30	3000.00	Pending

4. Budget Structure (BST)

Budget ID	Fiscal Year	Fund Code	Department Code	Original Budget	Adjustments	Available Budget
BUD2024A	2024	GF01	HR01	50000.00	0.00	50000.00
BUD2024B	2024	GF02	FIN02	75000.00	5000.00	80000.00
BUD2024C	2024	GF03	IT03	100000.00	-10000.00	90000.00
BUD2024D	2024	GF01	HR01	30000.00	0.00	30000.00

5. Cash Receipt (CR)

Receipt ID	Customer ID	Receipt Date	Payment Amount	Payment Method	Status
CR1001	CUST001	2024-01-05	1500.00	Check	Applied
CR1002	CUST002	2024-01-10	2000.00	ACH	Pending
CR1003	CUST003	2024-01-12	1800.00	Credit Card	Applied
CR1004	CUST004	2024-01-15	3000.00	Check	Rejected

Module : Inventory and Relationship Matrix



1. Financial Management Module

Component	Type	Key Tables/Transactions	Dependencies/Relationships	Baseline Artifacts/Data
General Ledger (GL)	Core Module	GL Account (GAC), GL Journal (GLJ)	Relies on Chart of Accounts (COA), Fund Codes, Department Codes	COA setup, Fiscal Year, Fund Codes
Accounts Payable (AP)	Submodule	AP Invoice (API), Vendor (VEN)	Linked to Vendor Management, Cash Management	Vendor Profiles, Payment Terms
Accounts Receivable (AR)	Submodule	AR Invoice (ARI), AR Payment (ARP)	Interfaces with Revenue Management	Customer Profiles, Payment Methods
Budgeting	Submodule	Budget Structure (BST), Budget Entry (BJE)	Relies on Fiscal Periods, GL Account Hierarchy	Budget Profiles, Budget Control Setup
Asset Management	Submodule	Asset Master (ASM), Depreciation (DEP)	Links to GL for asset transactions and depreciation	Asset Categories, Depreciation Policies

2. Procurement and Vendor Management Module

Component	Type	Key Tables/Transactions	Dependencies/Relationships	Baseline Artifacts/Data
Vendor Management (VCC)	Core Module	Vendor Profile (VEN), Vendor Address (VAD)	Used by AP, Procurement	Vendor Types, Vendor Addresses
Purchase Orders (PO)	Transaction	Purchase Order (POD), PO Line (POL)	Linked to Budgeting, AP for encumbrances	Item Catalog, PO Approval Workflow
Contract Management (CM)	Submodule	Contract (CON), Contract Amendment (CAM)	Links to Vendor, Procurement, Budgeting	Contract Types, Contract Templates

Component	Type	Key Tables/Transactions	Dependencies/Relationships	Baseline Artifacts/Data
Sourcing and Bids	Submodule	Bid Solicitation (BID), Bid Response (BIR)	Relates to Vendor, PO, and Contract	Bid Evaluation Criteria, Bidder Information

3. Human Resources and Payroll Module

Component	Type	Key Tables/Transactions	Dependencies/Relationships	Baseline Artifacts/Data
Employee Management	Core Module	Employee Profile (EMP), Employee Address (EAD)	Integrated with Payroll, Benefits, and Training	Employee Roles, Job Codes, Pay Grades
Payroll	Submodule	Payroll Run (PRR), Pay Calculation (PAY)	Relies on Employee, Job, Time and Attendance	Payroll Calendar, Tax Withholding Tables
Benefits Administration	Submodule	Benefits Plan (BEN), Enrollment (ENR)	Linked to Employee and Payroll	Benefits Plans, Employee Benefits Eligibility
Time and Attendance	Submodule	Time Entry (TME), Leave Request (LVR)	Connects to Payroll, Employee for attendance tracking	Time Codes, Leave Policies

4. Revenue and Accounts Receivable Module

Component	Type	Key Tables/Transactions	Dependencies/Relationships	Baseline Artifacts/Data
Revenue Entry (RE)	Core Module	Revenue Entry (RE), Revenue Line (REL)	Links to AR, Customer Billing Profiles	Customer Profiles, Revenue Codes
Customer Billing Profiles	Transaction	Billing Profile (BP), Customer Account (CA)	Used by AR for third-party billing	Customer Account Types, Payment Terms
Cash Receipts	Submodule	Cash Receipt (CR), Payment Application (PA)	Relies on AR, Banking for cash management	Payment Methods, Bank Accounts

5. Budget Management Module

Component	Type	Key Tables/Transactions	Dependencies/Relationships	Baseline Artifacts/Data
Budget Structure (BS)	Core Module	Budget Control (BC), Budget Allocation (BA)	Related to GL, used by all modules requiring budget information	Budget Structure Codes, Budget Levels

Component	Type	Key Tables/Transactions	Dependencies/Relationships	Baseline Artifacts/Data
Budget Adjustment	Transaction	Budget Adjustment Entry (BAE)	Adjustments impact GL, Procurement, and other budget-reliant modules	Budget Transfer Rules, Budget Approval Workflow

6. Reporting and Inquiry Tools

Component	Type	Key Reports/Inquiries	Dependencies/Relationships	Baseline Artifacts/Data
Financial Reports	Core Module	Trial Balance, Income Statement	Linked to GL, AR, AP	Reporting Periods, GL Account Groups
Budget Reports	Core Module	Budget Inquiry (BI), Available Budget Report (ABR)	Relates to Budget Structure, Control Tables	Budget Control Settings, Fiscal Periods
Procurement Reports	Core Module	PO Status, Contract Usage	Linked to Vendor, Contract, and PO	Report Templates, Approval Workflow Definitions

7. System Administration and Security

Component	Type	Key Tables/Configurations	Dependencies/Relationships	Baseline Artifacts/Data
User Management	Core Module	User Profile (USR), Role Assignment (ROL)	Linked to module access and transaction security	Role Definitions, Access Control Rules
Workflow and Approvals	Configuration	Workflow Setup (WFL), Approval Routing (APR)	Workflows define approval paths for financial transactions	Workflow Rules, Approval Levels
System Configuration	Core Module	System Options (SYS), Control Codes (CC)	Control settings across modules for consistent configurations	Fiscal Year Setup, Module-Specific Parameters

Relationship Matrix Overview

Modules	Primary Transactions and Tables	Dependent Components
General Ledger (GL)	GL Account (GAC), GL Journal (GLJ)	Accounts Payable, Revenue Entry, Asset Management, Reporting
Accounts Payable (AP)	AP Invoice (API), Vendor (VEN)	GL for expenditure tracking, Vendor Management
Accounts Receivable (AR)	AR Invoice (ARI), AR Payment (ARP)	Linked to Customer, Revenue Entry for collection purposes

Modules	Primary Transactions and Tables	Dependent Components
Budgeting	Budget Structure (BST), Budget Control (BC)	Impacts GL, Procurement, Revenue transactions
Procurement	Purchase Order (POD), Vendor Management (VCC)	Budget Control, AP for encumbrance management
Revenue	Revenue Entry (RE), Customer Profile (CP)	Linked to Budgeting, AR, and Customer Billing Profiles
Payroll	Payroll Run (PRR), Employee Profile (EMP)	Dependent on HR employee setup, Budgeting for payroll expense

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Module 12: Creating a CR (Cash Receipt) transaction in CGI Advantage 4 that references an RE (Receivable)



Overview

This guide provides a step-by-step process for creating a **CR (Cash Receipt)** transaction in CGI Advantage 4 that references an **RE (Receivable Entry)** event. Follow these instructions to apply payments against receivable entries accurately.

Step 1: Verify the RE (Receivable Entry) Event

1. Navigate to Receivable Entry (RE)

- Go to **Financials > Revenue > Receivable Entry (RE)**.

2. Search for the RE Transaction

- Enter relevant search criteria, such as **Transaction ID**, **Customer ID**, or **Date**.
- Click **Search** to locate the RE transaction you want to reference.

3. Verify the RE Details

- Open the RE transaction and verify details like **Customer ID**, **Amount Due**, **Outstanding Balance**, and **Payment Terms**.
- Note the **Transaction ID** and **Line Item IDs** (if applicable) for later reference in the CR transaction.

Step 2: Create the Cash Receipt (CR) Transaction

1. Navigate to Cash Receipt (CR)

- Go to **Financials > Cash Management > Cash Receipt (CR)**.

2. Create New CR Transaction

- Click **Create New** to start a new Cash Receipt transaction.

3. Enter Header Information

- **Transaction Date:** Enter the date the payment was received.
- **Receipt Number:** Assign a unique receipt number, or use the system-generated one if available.
- **Customer ID:** Enter the **Customer ID** that matches the one used in the RE transaction.
- **Payment Method:** Choose the payment method, such as **Check**, **ACH**, or **Cash**.
- **Amount Received:** Enter the total amount received from the customer.
- **Bank Account:** Select the bank account where the funds will be deposited.

Step 3: Reference the RE Transaction in CR Line Items

1. Add Line Item for the Cash Receipt

- In the **Line Items** section, click **Add Line Item**.

2. Enter Line Item Details

- **Receivable Event ID:** Enter the **Transaction ID** of the RE event that this payment will apply to.
- **Line Item ID (if applicable):** If the RE transaction includes multiple line items, specify the exact **Line Item ID** to which this payment should apply.
- **Amount Applied:** Enter the amount to be applied to the RE event. This amount can match the outstanding balance or be a partial payment.

3. Add Additional Line Items (if needed)

- If the payment applies to multiple RE events or line items, repeat the process for each line item by referencing the appropriate **Receivable Event ID** and **Line Item ID**.

Step 4: Review and Validate the Cash Receipt Transaction

1. Validate the Cash Receipt (CR) Entry

- Ensure that the **Amount Received** matches the total of all line items.
- Double-check that each **Receivable Event ID** and **Amount Applied** aligns with the correct RE transactions.

2. Check for Errors or Warnings

- CGI Advantage 4 will display any validation errors or warnings. Address these before proceeding.

Step 5: Submit or Approve the CR Transaction

1. Save the Transaction

- Click **Save** to save the transaction details.

2. Submit the Cash Receipt (CR) for Approval

- If your organization requires approval for cash receipts, submit the CR transaction by clicking **Submit**.
- The transaction will then be routed through the approval workflow as per your organization's configuration.

3. Finalize the Transaction

- Once approved (if required), the CR transaction will be posted, and the payment will be applied against the RE receivable, reducing the outstanding balance on the customer's account.

Step 6: Verify Application of Payment

1. Check the RE Transaction Status

- After posting, return to the **Receivable Entry (RE)** module.
- Search for the original RE transaction and open it to verify that the payment has been applied correctly.

2. Confirm Updated Balance

- Review the **Outstanding Balance** on the RE transaction. The applied payment should reduce this balance accordingly.

3. Generate or Print Receipt (if needed)

- If your organization requires a printed receipt, use the **Print Receipt** option in the CR transaction or generate a report to send to the customer.

Additional Notes

- **Partial Payments:** If a partial payment is applied, the outstanding balance on the RE transaction will reflect the remaining amount due.
- **Overpayments:** If the payment exceeds the amount due, the system may generate a credit balance or an adjustment, depending on configuration.

- **Audit and Review:** Ensure that all CR and RE transactions align with the customer's account details for accurate reporting and audit purposes.

Steps to Create an LCRR (Labor Cost Redistribution Request) in CGI Advantage 4

Overview

The **LCRR (Labor Cost Redistribution Request)** transaction is used to reallocate labor costs from one account or project to another. This guide walks you through the steps to create, validate, and submit an LCRR transaction in CGI Advantage 4.

Steps to Create an LCRR

1. Access the LCRR Transaction Page

1. Navigate to **Financials > Labor Management > Labor Cost Redistribution Request (LCRR)**.
2. Click on **Create New** to open a new LCRR transaction.

2. Enter Header Information

1. Transaction Code:

- Select the appropriate transaction code for the redistribution request, e.g., **LCRR**.

2. Transaction Date:

- Enter the date the transaction will take effect.
- This should align with the accounting period for the redistribution.

3. Description:

- Provide a brief description of the redistribution request (e.g., "Reallocate labor costs for Q1").

4. Reason Code:

- Choose a reason code that explains the purpose of the redistribution (e.g., **Correction**, **Budget Adjustment**, etc.).

3. Enter Line Item Details

1. Source Line Items:

- Click on **Add Line** to specify the original cost(s) to be redistributed.
- **Fields to Complete:**
 - **Account Code:** Enter the account code from which labor costs are being moved.
 - **Fund Code:** Enter the associated fund code for the original account.
 - **Amount:** Specify the amount to be redistributed (can be a partial or full amount).
 - **Reference Employee ID:** Enter the employee ID whose labor costs are being adjusted.

2. Target Line Items:

- Click on **Add Line** to specify where the labor costs are being moved.

- **Fields to Complete:**
- **Account Code:** Enter the account code to which the labor costs are being moved.
- **Fund Code:** Enter the associated fund code for the target account.
- **Amount:** Specify the redistributed amount.
- **Reference Employee ID:** Enter the same employee ID to maintain labor tracking.

4. Validate the Transaction

1. Run Validation:

- Click on the **Validate** button to ensure that all fields are correctly entered and comply with system rules.

2. Resolve Errors:

- If any validation errors appear, correct them as specified (e.g., mismatched funds, unavailable budget).

5. Submit the LCRR for Approval

1. Review:

- Carefully review the transaction details, including all source and target line items, to ensure accuracy.

2. Submit:

- Click on **Submit** to route the LCRR transaction through the approval workflow.
- The system will assign the transaction to the appropriate approver(s) based on configured workflows.

6. Monitor the Approval Process

1. Check Status:

- Navigate to the **Transaction Inquiry** page to monitor the status of your LCRR transaction.
- Use the **Transaction ID** to search for updates.

2. Approval Workflow:

- If approvals are required, follow up with approvers to ensure timely processing.

7. Finalize the Transaction

1. Post the Transaction:

- Once approved, the LCRR transaction will be posted automatically or manually, depending on your system configuration.

2. Verify Redistribution:

- Use reporting tools such as the **Labor Redistribution Report** or the **General Ledger Inquiry** to confirm that the labor costs were correctly reallocated.

Additional Notes

- **Budget Compliance:**
 - Ensure that the target accounts have sufficient budget availability before submitting the transaction.
- **Audit Trail:**
 - The LCRR transaction includes an audit trail to track changes and approvals. Ensure all comments and justifications are entered in the system for future reference.
- **Recurring Redistributions:**
 - For frequent redistributions, consider using templates or predefined entries to save time.

Summary

The LCRR transaction in CGI Advantage 4 enables precise reallocation of labor costs, ensuring accurate accounting and budget alignment. Following these steps will help you create, validate, and process LCRR transactions efficiently while maintaining compliance with organizational policies.

Module 15: Comprehensive Guide to End-User Pages, Fields, Screens, and Functions in CGI Advantage 4



Overview

This guide outlines the baseline, end-user-facing pages, fields, and screens in CGI Advantage 4. It is structured by module, focusing on **Financial Management (FIN)** and **Human Resources Management (HRM)** functionalities. Each section details key pages, their purpose, and the user-interactive fields available.

1. System Overview and Navigation

1.1 Main Navigation

- **Purpose:** Provides access to all modules, transactions, and reports.
- **Features:**
 - **Modules:** Organized into categories such as Financials, HRM, Procurement, and Reporting.
 - **Global Search:** Quickly locate pages, transactions, or records using keywords or transaction IDs.
 - **Favorites:** Users can bookmark frequently accessed pages.
- **Key User Actions:**
 - Navigate via menu paths or use the search bar.
 - Add pages to favorites for quick access.

1.2 Dashboard

- **Purpose:** Displays a personalized overview of user-specific tasks and key metrics.

- **Widgets:**
 - Pending Approvals
 - Notifications
 - Key Financial Metrics (e.g., budgets, revenue)
 - **User Actions:**
 - Configure dashboard widgets.
 - Click notifications to navigate directly to the relevant page.
-

2. Financial Management (FIN) Module

2.1 General Ledger (GL)

Key Pages and Screens

- **Journal Entry (GLJ):**
- **Purpose:** Create, edit, and post journal entries.
- **Fields:**
- Journal ID
- Fiscal Period
- Account Code
- Debit Amount
- Credit Amount
- **Actions:**
- Validate entry, Save draft, Submit for approval.
- **Trial Balance Inquiry:**
- **Purpose:** View account balances for selected periods.
- **Fields:**
- Fiscal Year
- Period Range
- Fund

- **Actions:**

- Generate Trial Balance, Export to Excel.

Screens

- **GL Dashboard:**

- Highlights open journal entries, posted transactions, and period-end tasks.

2.2 Accounts Payable (AP)

Key Pages and Screens

- **AP Invoice Entry (API):**

- **Purpose:** Enter vendor invoices for payment processing.

- **Fields:**

- Invoice Number
- Vendor ID
- Invoice Date
- Due Date
- Payment Terms
- Amount

- **Actions:**

- Validate invoice, Submit for approval, Post.

- **AP Payment (APP):**

- **Purpose:** Process payments for approved invoices.

- **Fields:**

- Payment ID
- Payment Method
- Vendor ID
- Amount

- **Actions:**

- Generate payment, Print check.

Screens

- **AP Dashboard:**

- Displays pending invoices, payment batches, and vendor balances.

2.3 Accounts Receivable (AR)

Key Pages and Screens

- **Receivable Entry (RE):**

- **Purpose:** Record revenue transactions, including customer billing.

- **Fields:**

- Receivable ID
- Customer ID
- Revenue Code
- Amount

- **Actions:**

- Save draft, Validate, Submit.

- **Cash Receipt (CR):**

- **Purpose:** Apply customer payments to outstanding receivables.

- **Fields:**

- Receipt ID
- Payment Method
- Amount Received
- Applied Amount

- **Actions:**

- Apply payment, Post.

Screens

- **AR Dashboard:**

- Shows outstanding receivables, payments applied, and aged receivables.

2.4 Budgeting

Key Pages and Screens

- **Budget Entry (BJE):**

- **Purpose:** Enter and adjust budget allocations.

- **Fields:**

- Budget ID

- Fiscal Year
- Department
- Original Budget
- Adjustments
- **Actions:**
- Validate, Submit for approval.
- **Budget Control (BC):**
- **Purpose:** Monitor budget compliance and spending limits.
- **Fields:**
- Control Group
- Fund Code
- Budget Balance
- **Actions:**
- View budget compliance reports.

Screens

- **Budget Dashboard:**
 - Displays budget summary, overages, and unspent funds.
-

3. Human Resources Management (HRM) Module

3.1 Employee Management

Key Pages and Screens

- **Employee Profile (EMP):**
- **Purpose:** Maintain employee records.
- **Fields:**
- Employee ID
- Name

- Department
- Job Title
- Status
- **Actions:**
- Create, Edit, Terminate employee.
- **Employee Address:**
- **Purpose:** Record employee addresses.
- **Fields:**
- Address Type (Home, Work)
- Address Line 1
- City, State, Zip Code
- **Actions:**
- Add, Update, Delete.

Screens

- **Employee Dashboard:**
- Summarizes employee counts, pending actions, and recent updates.

3.2 Payroll Processing

Key Pages and Screens

- **Payroll Run (PRR):**
- **Purpose:** Process employee payroll.
- **Fields:**
- Pay Period Start
- Pay Period End
- Gross Pay
- Deductions
- Net Pay
- **Actions:**
- Calculate, Validate, Submit.
- **Pay History:**

- **Purpose:** View historical payroll data for employees.
- **Fields:**
 - Employee ID
 - Pay Period
 - Gross Pay
- **Actions:**
 - Generate reports.

Screens

- **Payroll Dashboard:**
 - Displays payroll runs, errors, and pending approvals.

3.3 Benefits Administration

Key Pages and Screens

- **Benefit Plan Entry:**
 - **Purpose:** Define benefits plans available to employees.
 - **Fields:**
 - Plan ID
 - Description
 - Coverage Type
 - Premium Amount
 - **Actions:**
 - Add, Edit, Delete.
- **Benefits Enrollment:**
 - **Purpose:** Enroll employees in benefit plans.
 - **Fields:**
 - Employee ID
 - Benefit Plan ID
 - Coverage Level
 - **Actions:**
 - Enroll, Update, Terminate.

Screens

- **Benefits Dashboard:**
 - Summarizes plan enrollments, coverage levels, and premium totals.
-

4. Workflow and Approvals

Key Pages and Screens

- **Approval Queue:**
 - **Purpose:** Manage and process approvals.
 - **Fields:**
 - Transaction ID
 - Approver Name
 - Status
 - **Actions:**
 - Approve, Reject, Forward.
 - **Workflow Configuration:**
 - **Purpose:** Define approval workflows.
 - **Fields:**
 - Workflow ID
 - Steps
 - Rules
 - **Actions:**
 - Activate, Edit, Deactivate.
-

Summary

This guide provides an end-to-end overview of the baseline, end-user-facing pages, fields, screens, and functionalities in CGI Advantage 4. Use it as a reference for system training, operational onboarding, or troubleshooting common user tasks.

Module 19: CGI Advantage 4 Data Setup for RE Transaction with Multiple Third-Party Billing Addresses



Overview

This dataset covers the necessary tables and dependencies to configure an RE transaction where a primary customer receives services, but billing is directed to multiple third-party customers, each with their own billing address. The setup includes customer profiles, billing profiles, billing rate tables, and other related tables to ensure that all dependencies are met.

1. Vendor Customer Creation (VCC) - Customer Profile

Customer ID	Customer Name	Customer Type	Tax ID	Default Billing Address	Contact Info
CUST001	Main School District	Public Entity	99-9999999	ADDR01	main@schools.org
CUST002	Grant Organization 1	Third-Party Org	55-5555555	ADDR02	billing@grant1.org
CUST003	Grant Organization 2	Third-Party Org	44-4444444	ADDR03	billing@grant2.org
CUST004	Sponsoring Foundation	Third-Party Org	33-3333333	ADDR04	billing@foundation.org

2. Vendor Customer Address (VAD) - Customer Address Information

Address ID	Customer ID	Address Line 1	City	State	Zip Code
ADDR01	CUST001	123 Education St.	Springfield	IL	62701
ADDR02	CUST002	456 Grant Blvd.	Big City	NY	10010
ADDR03	CUST003	789 Sponsorship Rd.	Metropolis	CA	90001
ADDR04	CUST004	321 Foundation Ave.	Capital City	DC	20001

3. Billing Profile Table (BP) - Setting Up Billing Profiles for Multiple Third-Party Customers

Billing Profile ID	Customer ID	Third-Party Customer	Address ID	Payment Terms	Rate Table ID
BP001	CUST001	CUST002	ADDR02	Net 30	RATE01
BP002	CUST001	CUST003	ADDR03	Net 45	RATE02
BP003	CUST001	CUST004	ADDR04	Net 60	RATE03

4. Customer Account Options Table (CACT) - Linking Customers to Third-Party Billing Options

Account Option ID	Primary Customer	Third-Party Customer	Address ID	Billing Instructions
CACT01	CUST001	CUST002	ADDR02	Bill Grant Org 1 for tuition services
CACT02	CUST001	CUST003	ADDR03	Bill Grant Org 2 for training sessions
CACT03	CUST001	CUST004	ADDR04	Bill Foundation for equipment rental

5. Billing Rate Table (BRT) - Defining Rates Based on Services and Third-Party Customers

Rate Table ID	Service Description	Rate Code	Rate Amount	Unit of Measure	Effective Date
RATE01	Tuition per student	RT01	500.00	Student	2024-01-01
RATE02	Training Session	RT02	2000.00	Session	2024-01-01
RATE03	Equipment Rental	RT03	150.00	Day	2024-01-01

6. Revenue Codes Table - Defining Revenue Codes for Different Services

Revenue Code	Revenue Description	Account Code
REV001	Tuition Revenue	4000
REV002	Training Revenue	4100
REV003	Equipment Rental	4200

7. General Ledger Account (GAC) - Linking Accounts for Revenue Recording

Account ID	Account Name	Account Type	Department Code
4000	Tuition Revenue	Revenue	EDU01
4100	Training Revenue	Revenue	EDU01
4200	Equipment Rental	Revenue	EDU01

8. Chart of Accounts (COA) - Department and Fund Setup

Department Code	Fund Code	Account Code	Account Description
EDU01	GF01	4000	Tuition Revenue for Education Department
EDU01	GF01	4100	Training Revenue for Education Department
EDU01	GF01	4200	Equipment Rental Revenue for Education Department

9. Fiscal Period Table - Setting Up Fiscal Periods

Fiscal Year	Period ID	Start Date	End Date
2024	01	2024-01-01	2024-01-31
2024	02	2024-02-01	2024-02-29
2024	03	2024-03-01	2024-03-31

10. Receivable Entry (RE) - Example RE Transaction with Multiple Third-Party Billing

Transaction ID	Customer ID	Billing Profile ID	Revenue Code	Quantity	Description
RE1001	CUST001	BP001	REV001	10	Tuition billing for Grant Org 1
RE1002	CUST001	BP002	REV002	1	Training session for Grant Org 2
RE1003	CUST001	BP003	REV003	5	Equipment rental for Sponsoring Foundation

Process Overview

1. **Set Up Customer Profiles:** Create profiles for the primary customer and all third-party customers involved in the RE transaction.
2. **Define Billing Addresses:** Set up addresses for each third-party customer in the VAD table to ensure proper billing locations.
3. **Create Billing Profiles:** Define separate billing profiles in the Billing Profile Table, linking each third-party customer to the primary customer.
4. **Establish Customer Account Options:** Use CACT entries to specify third-party billing relationships and instructions.
5. **Set Rates in Billing Rate Table:** Define service rates and effective dates for billing calculations.
6. **Generate RE Transaction:** In the RE table, assign the billing profile ID for each line item to bill the correct third-party payer on behalf of the primary customer.

Notes

- Ensure that each billing profile in the Billing Profile Table references the correct third-party customer and address ID to prevent billing errors.
- Fiscal periods and account structures must be verified to ensure transactions align with reporting requirements.
- The Billing Rate Table should reflect updated rates to avoid miscalculations in the RE transaction.

Module 23: Sample Data for FIN and HRM Modules in CGI Advantage 4



Financial Management (FIN) Module

1. General Ledger Accounts (GL_ACCOUNT)

Account ID	Account Description	Account Type	Department Code
4000	Tuition Revenue	Revenue	EDU01
5000	Salaries Expense	Expense	HR01
6000	Office Supplies Expense	Expense	FIN02
7000	IT Equipment Expense	Expense	IT03
8000	Training Budget	Expense	HR01

2. Vendors (VENDOR_PROFILE)

Vendor ID	Vendor Name	Payment Terms	Contact Info
VEND001	ABC Office Supplies Co.	Net 30	contact@abcoffice.com
VEND002	XYZ Consulting Services	Net 45	billing@xyzconsult.com
VEND003	Global IT Solutions	Net 15	it@globalitsolutions.com
VEND004	Training Excellence Inc.	Net 60	info@trainingexcellence.com

3. Customers (CUSTOMER_PROFILE)

Customer ID	Customer Name	Contact Info
CUST001	Springfield School District	billing@schools.org
CUST002	Grant Foundation	finance@grantfoundation.org
CUST003	Education Sponsorship Fund	payments@educationsponsors.org

4. Budget Structures (BUDGET_STRUCTURE)

Budget ID	Fiscal Year	Fund Code	Department Code	Original Budget
BUD2024A	2024	GF01	HR01	500000.00
BUD2024B	2024	GF02	FIN02	250000.00
BUD2024C	2024	GF03	IT03	150000.00

5. Purchase Orders (PURCHASE_ORDER)

PO Number	Vendor Name	Date	Status	Total Amount
PO1001	ABC Office Supplies Co.	2024-01-10	Approved	2000.00
PO1002	XYZ Consulting Services	2024-01-12	Approved	5000.00
PO1003	Global IT Solutions	2024-01-15	Pending	7500.00
PO1004	Training Excellence Inc.	2024-01-18	Rejected	3000.00

6. Invoices (AP_INVOICE)

Invoice ID	Vendor Name	Date	Status	Amount Due
INV1001	ABC Office Supplies Co.	2024-01-05	Paid	1000.00
INV1002	XYZ Consulting Services	2024-01-07	Pending	2500.00
INV1003	Global IT Solutions	2024-01-10	Approved	1500.00

7. Payments (AP_PAYMENT)

Payment ID	Invoice ID	Date	Amount Paid
PAY1001	INV1001	2024-01-15	1000.00
PAY1002	INV1002	2024-01-20	2500.00
PAY1003	INV1003	2024-01-25	1500.00

8. Journal Entries (GL_JOURNAL)

Journal ID	Date	Description	Total Amount
JE1001	2024-01-03	Office Supplies Expense Entry	2000.00
JE1002	2024-01-06	Consulting Services Payment	5000.00
JE1003	2024-01-08	IT Equipment Purchase	7500.00

Human Resources Management (HRM) Module

1. Employee Profiles (EMPLOYEE_PROFILE)

Employee ID	First Name	Last Name	Department Code	Job Title
1001	John	Doe	HR01	HR Manager

Employee ID	First Name	Last Name	Department Code	Job Title
1002	Jane	Smith	FIN02	Finance Analyst
1003	Alice	Johnson	IT03	IT Specialist
1004	Bob	Lee	HR01	HR Assistant

2. Job Titles (JOB_TITLES)

Job Code	Job Title
HRMGR	HR Manager
FINAN	Finance Analyst
ITSPC	IT Specialist
HRASST	HR Assistant

3. Payroll Data (PAY_CALCULATION)

Employee ID	Pay Period Start	Pay Period End	Gross Pay	Deductions
Net Pay	1001	2024-01-01	2024-01-15	3000.00
300.00	2700.00	1002	2024-01-01	2024-01-15
4000.00	400.00	3600.00	1003	2024-01-01
2024-01-15	3500.00	350.00	3150.00	1004
2024-01-01	2024-01-15	2000.00	200.00	1800.00

4. Benefits Plans (BENEFIT_PLAN)

Benefit Plan ID	Plan Description	Premium
MED01	Medical Insurance	200.00
DEN01	Dental Insurance	50.00
VSN01	Vision Insurance	20.00

5. Benefits Enrollment (BENEFIT_ENROLLMENT)

Employee ID	Benefit Plan ID	Coverage Level	Enrollment Date
1001	MED01	Family	2024-01-01
1002	DEN01	Individual	2024-01-01
1003	VSN01	Family	2024-01-01
1004	MED01	Individual	2024-01-01

6. Time Entries (TIME_ENTRY)

Time Entry ID	Employee ID	Date	Hours Worked	Time Code
TE10001	1001	2024-01-10	8	REG
TE10002	1002	2024-01-11	8	REG
TE10003	1003	2024-01-12	6	SICK

Time Entry ID	Employee ID	Date	Hours Worked	Time Code
TE10004	1004	2024-01-13	4	VAC

7. Leave Balances (LEAVE_BALANCE)

Employee ID	Leave Type	Accrued Leave	Used Leave
1001	Vacation	80	8
1002	Sick	40	6
1003	Vacation	60	4
1004	Sick	30	2

Module 21: Requirements and Dependencies for Creating and Processing an RE Receivable Transaction Using the Billing Rate Table in CGI Advantage 4



Overview

To create and process an **RE (Receivable Entry) transaction** that uses the **Billing Rate Table** in CGI Advantage 4, there are several system requirements and dependencies. These dependencies ensure that billing rates are applied accurately based on predefined tables, customer profiles, and account structures. This document outlines the requirements, prerequisites, and setup dependencies necessary for successful RE transaction processing.

Requirements and Dependencies

Requirement/Dependency	Description
Customer Profile Setup	A customer profile must be created in the Vendor Customer Creation (VCC) module. This profile includes essential customer information such as name, address, Tax ID, and any billing relationships or payment terms specific to the customer.
Billing Profile Table	The Billing Profile Table must contain a record for each customer requiring third-party billing or special rate applications. The profile should specify the rate schedule and, if applicable, the address ID associated with each customer's billing details.
Billing Rate Table	The Billing Rate Table must be configured with rate codes, rate types, and applicable rate amounts. This table determines the billing rate calculations applied to the RE transaction based on service, customer, or other criteria.
Chart of Accounts (COA)	The Chart of Accounts must be set up with the necessary account codes that correspond to the revenue accounts being billed. This ensures that the receivable entries are allocated correctly within the general ledger.

Requirement/Dependency	Description
Revenue Codes	Revenue codes must be created and linked to specific accounts in the Revenue Module . Each revenue code represents a distinct income source and determines how receivable transactions are recorded in the general ledger.
Budget Control and Structure	The budget structure and control settings should be defined for accounts impacted by RE transactions. This may involve setting up budget controls and ensuring that funds are available for the services billed.
Service or Product Codes	If billing is based on specific services or products, these codes must be established and linked to rates in the Billing Rate Table. Each service code should map to a rate and, if applicable, an account code for tracking revenue accurately.
Customer Account Options (CACT Table)	The Customer Account Options Table (CACT) must define specific billing options for each customer. If Address ID inference is required, the CACT table should include entries that link customers to particular billing addresses and preferences.
User Access and Permissions	Users must have appropriate access rights to create and process RE transactions, view the Billing Rate Table, and access the Billing Profile Table. Role-based permissions should be configured in User Management .
Fiscal Period Setup	The fiscal periods must be defined for the current fiscal year. This ensures that all transactions are recorded in the correct accounting period and that revenue can be tracked accurately over time.
Approval Workflow Configuration	If approvals are required for RE transactions, the approval workflow must be configured to route RE transactions to designated approvers. The workflow should be established in Workflow and Approvals settings.
Tax Codes (if applicable)	If taxes are applied to the receivable transactions, tax codes must be defined. Tax codes should align with the relevant billing rate and service/product codes.
Billing Frequency and Payment Terms	Billing frequency (e.g., monthly, quarterly) and payment terms should be set up within the Billing Profile associated with each customer. This helps determine the timing and terms under which the customer will be billed and expected to pay.
System Configuration Settings	System settings must allow for the use of the Billing Rate Table in RE transactions. This may include enabling specific modules or transaction types and ensuring that dependencies like Rate Codes are active.

Process Overview

Once all dependencies are met, the following steps outline the process for creating and processing an RE transaction with the Billing Rate Table:

- Create or Verify Customer Profile:** Ensure the customer profile exists in VCC with relevant billing information and payment terms.
- Set Up Billing Profile:** Define or verify the billing profile in the Billing Profile Table, specifying the Billing Rate Table reference.
- Create RE Transaction:**

- Go to **Revenue Entry (RE)** and start a new transaction.
 - Select the **Customer ID** and apply the **Billing Rate Table** for rate calculations.
 - Enter required line items, specifying revenue codes, service/product codes, and rate codes as needed.
4. **Verify Budget and Funds:** Use the Budget Inquiry to confirm available budget for accounts affected by the transaction.
 5. **Submit for Approval:** If approvals are configured, submit the RE transaction for routing through the established workflow.
 6. **Monitor and Complete Transaction:** Upon approval, the RE transaction is processed and recorded. Monitor transaction status to ensure completion.

Notes

- **Dependencies Verification:** Confirm all tables, profiles, and rate codes are updated and accurate before initiating an RE transaction.
- **Role Permissions:** Ensure users have the necessary access to the Billing Rate Table and related RE transaction components.
- **Rate Calculation:** Billing Rate Table settings determine the calculation method for applying rates in RE transactions. Verify that rate configurations align with organizational billing practices.

Requirements and Dependencies for Creating an RE Transaction for a Customer with Multiple Third-Party Billing Options in CGI Advantage 4

Overview

In CGI Advantage 4, you can create an **RE (Receivable Entry) transaction** for a customer who requires multiple third-party billing options. This allows a primary customer to receive services while designated third-party entities are billed on their behalf. This setup involves specific configuration in customer profiles, billing profiles, and address tables. This document outlines the requirements, dependencies, and configuration steps to process RE transactions under this arrangement.

Requirements and Dependencies

Requirement/Dependency	Description
Customer Profile Setup	Create or verify a profile for the primary customer in Vendor Customer Creation (VCC) . This profile represents the customer receiving the service and links to the billing profiles and options for multiple third-party payers.
Billing Profile Table	Define multiple billing profiles in the Billing Profile Table for the customer. Each billing profile specifies a unique third-party payer and links to a corresponding Address ID for the designated billing recipient.
Third-Party Customer Profiles	Create profiles for each third-party customer responsible for paying bills on behalf of the primary customer. Each third-party customer should have a record in VCC , including address and contact information.
Billing Rate Table (if applicable)	If billing is based on rates, configure the Billing Rate Table with rate codes and applicable rates. These rates should align with the service or product codes and may vary by third-party customer as needed.
Chart of Accounts (COA)	Set up the Chart of Accounts to define revenue accounts impacted by the RE transaction. This ensures the receivables are accurately recorded in the general ledger.
Revenue Codes	Create and assign revenue codes to specific accounts within the Revenue Module . Revenue codes represent distinct income sources and will be referenced in the RE transaction.
Service or Product Codes	If billing varies by service or product, these codes must be established in the system and linked to the Billing Rate Table (if applicable). Each service or product code should map to an account code.

Requirement/Dependency	Description
Customer Account Options Table (CACT)	Use the Customer Account Options Table (CACT) to define options for multiple billing addresses and payers. Each third-party payer must have an entry in the CACT table that links the primary customer to specific billing preferences and addresses.
User Access and Permissions	Ensure users have appropriate permissions to create and process RE transactions, access the Billing Profile Table, and set up CACT configurations for multiple billing options. Permissions should be managed through User Management .
Fiscal Period Setup	Define fiscal periods for the current fiscal year to ensure transactions are recorded in the correct accounting period.
Approval Workflow Configuration	Configure approval workflows, if required, for RE transactions involving third-party billing. Approval routing should be set up in the Workflow and Approvals section.
Tax Codes (if applicable)	If taxes apply to the transaction, tax codes must be created in alignment with relevant service or product codes.
Payment Terms and Billing Frequency	Define payment terms and billing frequencies within each billing profile for third-party customers. This setup determines the terms under which third parties will be billed and expected to pay.
System Configuration Settings	Confirm that system settings enable third-party billing for RE transactions. Ensure that dependencies such as CACT and Billing Profile configurations are activated.

Process Overview

Once all dependencies are met, follow these steps to create and process an RE transaction with multiple third-party billing options:

- Create or Verify Customer Profile:** Ensure the primary customer’s profile exists in VCC with relevant billing information, and confirm each third-party payer profile is complete with contact and billing address details.
- Define Multiple Billing Profiles:** In the **Billing Profile Table**, create separate billing profiles for each third-party customer linked to the primary customer. Specify unique billing addresses using the Address ID linked to each third party in the CACT table.
- Link Third-Party Payers via CACT Table:** Set up entries in the **Customer Account Options Table (CACT)** for each third-party payer. This configuration allows the system to infer the correct Address ID and billing preferences when creating the RE transaction.
- Create RE Transaction:**
 - Navigate to **Revenue Entry (RE)** and initiate a new transaction.
 - Select the primary customer’s **Customer ID**.
 - Apply the appropriate billing profile for each line item that reflects the designated third-party payer.
 - Specify line item details such as revenue codes, service or product codes, and, if applicable, rate codes from the Billing Rate Table.

5. **Verify Budget and Available Funds:** Use Budget Inquiry to confirm available budget for accounts involved in the transaction.
6. **Submit for Approval:** If approval workflows are configured, submit the RE transaction to route it through the designated approvers.
7. **Monitor and Complete Transaction:** Upon approval, the RE transaction will be processed with each line item billed to the assigned third-party payer. Monitor transaction status for completion and invoice generation.

Notes

- **Billing Profile and CACT Linkage:** Ensure that each billing profile is linked accurately to the third-party customer's Address ID in the CACT table to avoid billing errors.
- **Role Permissions:** Verify that users have the necessary permissions to set up and view third-party billing profiles and access customer account options.
- **Rate Calculations:** When using the Billing Rate Table, confirm that rate configurations align with the service or product being billed and are appropriate for each third-party payer.

This configuration allows for flexible billing to multiple third-party customers on behalf of a primary customer, streamlining the receivables process in CGI Advantage 4.

Module 22: Requirements and Dependencies for Creating an RE Transaction for a Customer with Multiple Third-Party Billing Options in CGI Advantage 4



Overview

In CGI Advantage 4, you can create an **RE (Receivable Entry) transaction** for a customer who requires multiple third-party billing options. This allows a primary customer to receive services while designated third-party entities are billed on their behalf. This setup involves specific configuration in customer profiles, billing profiles, and address tables. This document outlines the requirements, dependencies, and configuration steps to process RE transactions under this arrangement.

Requirements and Dependencies

Requirement/Dependency	Description
Customer Profile Setup	Create or verify a profile for the primary customer in Vendor Customer Creation (VCC) . This profile represents the customer receiving the service and links to the billing profiles and options for multiple third-party payers.
Billing Profile Table	Define multiple billing profiles in the Billing Profile Table for the customer. Each billing profile specifies a unique third-party payer and links to a corresponding Address ID for the designated billing recipient.
Third-Party Customer Profiles	Create profiles for each third-party customer responsible for paying bills on behalf of the primary customer. Each third-party customer should have a record in VCC , including address and contact information.
Billing Rate Table (if applicable)	If billing is based on rates, configure the Billing Rate Table with rate codes and applicable rates. These rates should align with the service or product codes and may vary by third-party customer as needed.

Requirement/Dependency	Description
Chart of Accounts (COA)	Set up the Chart of Accounts to define revenue accounts impacted by the RE transaction. This ensures the receivables are accurately recorded in the general ledger.
Revenue Codes	Create and assign revenue codes to specific accounts within the Revenue Module . Revenue codes represent distinct income sources and will be referenced in the RE transaction.
Service or Product Codes	If billing varies by service or product, these codes must be established in the system and linked to the Billing Rate Table (if applicable). Each service or product code should map to an account code.
Customer Account Options Table (CACT)	Use the Customer Account Options Table (CACT) to define options for multiple billing addresses and payers. Each third-party payer must have an entry in the CACT table that links the primary customer to specific billing preferences and addresses.
User Access and Permissions	Ensure users have appropriate permissions to create and process RE transactions, access the Billing Profile Table, and set up CACT configurations for multiple billing options. Permissions should be managed through User Management .
Fiscal Period Setup	Define fiscal periods for the current fiscal year to ensure transactions are recorded in the correct accounting period.
Approval Workflow Configuration	Configure approval workflows, if required, for RE transactions involving third-party billing. Approval routing should be set up in the Workflow and Approvals section.
Tax Codes (if applicable)	If taxes apply to the transaction, tax codes must be created in alignment with relevant service or product codes.
Payment Terms and Billing Frequency	Define payment terms and billing frequencies within each billing profile for third-party customers. This setup determines the terms under which third parties will be billed and expected to pay.
System Configuration Settings	Confirm that system settings enable third-party billing for RE transactions. Ensure that dependencies such as CACT and Billing Profile configurations are activated.

Process Overview

Once all dependencies are met, follow these steps to create and process an RE transaction with multiple third-party billing options:

1. **Create or Verify Customer Profile:** Ensure the primary customer's profile exists in VCC with relevant billing information, and confirm each third-party payer profile is complete with contact and billing address details.
2. **Define Multiple Billing Profiles:** In the **Billing Profile Table**, create separate billing profiles for each third-party customer linked to the primary customer. Specify unique billing addresses using the Address ID linked to each third party in the CACT table.
3. **Link Third-Party Payers via CACT Table:** Set up entries in the **Customer Account Options Table (CACT)** for each third-party payer. This configuration allows the system to infer the correct Address ID and billing preferences when creating the RE transaction.
4. **Create RE Transaction:**

- Navigate to **Revenue Entry (RE)** and initiate a new transaction.
 - Select the primary customer's **Customer ID**.
 - Apply the appropriate billing profile for each line item that reflects the designated third-party payer.
 - Specify line item details such as revenue codes, service or product codes, and, if applicable, rate codes from the Billing Rate Table.
5. **Verify Budget and Available Funds:** Use Budget Inquiry to confirm available budget for accounts involved in the transaction.
 6. **Submit for Approval:** If approval workflows are configured, submit the RE transaction to route it through the designated approvers.
 7. **Monitor and Complete Transaction:** Upon approval, the RE transaction will be processed with each line item billed to the assigned third-party payer. Monitor transaction status for completion and invoice generation.

Notes

- **Billing Profile and CACT Linkage:** Ensure that each billing profile is linked accurately to the third-party customer's Address ID in the CACT table to avoid billing errors.
- **Role Permissions:** Verify that users have the necessary permissions to set up and view third-party billing profiles and access customer account options.
- **Rate Calculations:** When using the Billing Rate Table, confirm that rate configurations align with the service or product being billed and are appropriate for each third-party payer.

This configuration allows for flexible billing to multiple third-party customers on behalf of a primary customer, streamlining the receivables process in CGI Advantage 4.

Module : Requirements and Dependencies for Processing an RE Transaction Using the Billing Rate Table in CGI Advantage 4



To create and process a **Revenue Entry (RE)** transaction using the **Billing Rate Table** in CGI Advantage 4, ensure the following requirements and dependencies are configured. This setup allows the RE transaction to reference billing rates accurately, calculate charges, and complete the transaction workflow.

1. Set Up Customer and Billing Profiles

- **Customer Profile:** Ensure the customer receiving the service has an entry in the Vendor/Customer Creation (VCC) table.
 - Steps:
 - Ensure Customer ID, Customer Type, Tax ID, and Contact Information are complete.
- **Billing Profile:** Configure a Billing Profile for the customer in the Billing Profile Table.
 - Link the profile to the Customer ID.
 - Set the Billing Method to use the Billing Rate Table for rate-based billing.

2. Billing Rate Table Configuration

- **Billing Rate Table:** Define rates for services or items in the Billing Rate Table.
 - Rate Type: Define rate types (e.g., hourly, per unit).
 - Effective Dates: Set date ranges for each rate's validity.
 - Rate Amounts: Enter amounts based on service, product, or unit.
 - Unit of Measure: Specify per-hour, per-unit, etc.

- **Service Codes:** Associate service codes with rates to apply in RE transactions.
- **Validation:** Ensure no missing or conflicting entries to avoid transaction errors.

3. Chart of Accounts and Budget Control Setup

- **Chart of Accounts (COA):** Verify necessary accounts for revenue recognition, receivables, and costs.
- **Budget Controls:** Ensure budget availability for accounts used in RE transactions.
 - Confirm Budget Structure and levels are active.
 - Confirm appropriation for revenue accounts is available.

4. Customer Account Options and Address Configuration

- **Customer Account Options Table (CACT):** Set billing preferences like address inference and payment terms.
 - **Billing Address:** If Address ID needs to be inferred, configure in CACT.
 - **Payment Terms:** Set terms (e.g., Net 30, Net 60).
- **Multiple Address Support:** Configure multiple billing addresses and define Address Type in CACT or Customer Address table if needed.

5. Revenue Entry (RE) Transaction Configuration

- **Revenue Entry Form Access:** Ensure access to the RE form in the Revenue module.
- **Transaction Defaults:** Set up common field defaults:
 - Service Date, Customer ID, Service Codes, Units.
- **Calculation Rules:** Configure to ensure correct charge calculations from rate and unit data.

6. Workflow and Approval Setup

- **Workflow Configuration:** Set up approval workflows for RE transactions.

- **Role-Based Approvals:** Configure levels of authorization as needed.
- **Email Notifications:** Optionally, set notifications for approvers.

7. System Configuration and User Permissions

- **User Permissions:** Verify access for users involved in creating, processing, and approving RE transactions.
- **System Configuration:** Set parameters to support rate-based billing.
 - Fiscal Year Setup: Ensure the current period is open.
 - Currency Configuration: Set default currency if using multiple currencies.

8. Testing and Validation of Configuration

- **Test RE Transaction:** Run test transactions to verify calculations and references.
- **Error Checking:** Address any configuration issues flagged during testing.

Summary of Requirements

Requirement	Purpose
Customer Profile & Billing Profile	To define customer and billing preferences
Billing Rate Table	To establish rate structures for services or products
Chart of Accounts (COA)	For revenue recognition and cost tracking
Budget Control	Ensures availability of budget if necessary
Customer Account Options (CACT)	To define billing addresses and payment terms
RE Form Access and Defaults	To streamline RE transaction creation
Workflow and Approvals	For authorizing RE transactions
User Permissions	Grants access to necessary forms and transactions
System Configuration	Ensures parameters support billing rate processing
Testing and Validation	Confirms all configurations work as expected

Module 99: CGI Advantage 4 Crash Course for Developers



Overview

This crash course is designed for developers who are new to CGI Advantage 4. It provides an overview of core modules, architecture, development environment setup, common workflows, and customization methods. This guide serves as a quick reference to understand how to extend and maintain applications within CGI Advantage 4.

1. Understanding CGI Advantage 4 Architecture

CGI Advantage 4 is a comprehensive ERP system with a modular architecture that caters to government and public sector organizations. Core components include:

- **Modules:** CGI Advantage 4 has modules for Financial Management (FIN), Human Resources Management (HRM), Budgeting, Procurement, and more.
- **Frameworks:** Built on Java EE, CGI Advantage 4 utilizes JSP, servlets, and EJBs for backend processing.
- **Data Layer:** Typically integrated with Oracle databases, with tables and schemas customized for each client.
- **Customizations:** Configurable through scripts, templates, and XML, supporting custom business rules and workflows.

2. Setting Up the Development Environment

Follow these steps to set up the environment:

1. Install Required Software

- Java Development Kit (JDK) 1.8 or higher

- Apache Ant and Maven for build automation
- An Integrated Development Environment (IDE) such as Eclipse or IntelliJ IDEA
- Oracle SQL Developer or a similar database client for database management

2. Set Up Project Repositories

- Access the repository (e.g., Git, SVN) where your CGI Advantage project files are stored.
- Clone or check out the repository locally to set up your workspace.

3. Configure Database Connection

- Update database connection strings in the development configuration files, typically located in the `config` or `resources` directory.
- Test the database connection to ensure access to necessary tables and schemas.

4. Set Up Build Scripts

- Configure build scripts (Ant or Maven) to handle compilation, packaging, and deployment tasks.
- Common build targets include `clean`, `compile`, `package`, and `deploy`.

3. Key Modules and Tables

Developers should be familiar with these primary modules and tables:

Financial Management (FIN)

- **General Ledger (GL):** Core tables include `GL_ACCOUNT`, `GL_JOURNAL`.
- **Accounts Payable (AP):** Key tables are `AP_INVOICE`, `AP_PAYMENT`.
- **Accounts Receivable (AR):** Key tables include `AR_INVOICE`, `AR_PAYMENT`.
- **Budgeting:** Tables like `BUDGET_STRUCTURE` and `BUDGET_ENTRY`.

Human Resources Management (HRM)

- **Employee Profile:** Core tables include `EMPLOYEE_PROFILE`, `EMPLOYEE_ADDRESS`.
- **Payroll:** Tables like `PAYROLL_RUN` and `PAY_CALCULATION`.

Common Tables

- **Customer and Vendor:** Tables such as `CUSTOMER_PROFILE`, `VENDOR_PROFILE`.
- **Account Options:** `ACCOUNT_OPTIONS` table, linking accounts to configurations.

4. Core Development Tasks and Processes

Working with Pages and UI Components

CGI Advantage 4's UI is largely JSP-based, with HTML templates and CSS for styling.

- **Page Customizations:** Locate and modify JSP files within the `webapps` directory.
- **JavaScript Validation:** Use JavaScript files for field validations and UI enhancements.
- **CSS Styling:** Modify CSS files for custom themes and styles.

Customizing Business Rules

Business rules in CGI Advantage can be modified or created using XML configurations and Java classes.

- **Business Rule XMLs:** Found in `config` directories, these XML files define rule parameters.
- **Java Classes:** Extend or override business rules using Java classes by creating custom logic in service layer files.

Workflow and Approval Processes

- **Workflow Configuration:** Located in `workflow` XML files, where you can define routing and approval hierarchies.
- **Approval Process Customization:** Use the workflow engine to set rules and approvals by department, role, or transaction type.

5. Database Management and Table Customization

- **Creating Custom Tables:** Use SQL scripts to create tables, ensuring naming conventions are followed.
- **Modifying Core Tables:** Be cautious with core tables; ensure all changes are approved and follow CGI protocols.
- **Database Indexing:** Index frequently queried columns to improve performance.

6. Security and Permissions

- **Role-Based Access Control:** Define user roles and permissions using configuration files or

database tables like `USER_PROFILE` and `ROLE_ASSIGNMENT`.

- **Data Security:** Implement field-level encryption for sensitive data fields such as SSNs and financial data.
- **Audit Logging:** Enable audit trails for critical transactions and maintain logging for security reviews.

7. Common Development Scenarios

Scenario 1: Adding a Custom Field to a Page

1. **Add Column to Database:** Alter the relevant table to include a new column.
2. **Update JSP Page:** Modify the JSP file to include the new field.
3. **Configure Validation:** Add JavaScript validation as needed.
4. **Test and Deploy:** Test thoroughly to ensure the field displays and saves correctly.

Scenario 2: Modifying an Approval Workflow

1. **Open Workflow XML:** Locate the XML file for the desired workflow.
2. **Edit Routing Rules:** Adjust routing conditions or add new approval steps.
3. **Deploy and Test:** Ensure the workflow routes correctly through the approval chain.

Scenario 3: Creating a Report

1. **Define SQL Query:** Write an SQL query to pull data from relevant tables.
2. **Create Report Template:** Develop the report layout using JasperReports or another reporting tool.
3. **Integrate and Test:** Add the report to CGI Advantage and verify output.

8. Best Practices for Development

- **Version Control:** Use version control for all changes, committing frequently and writing clear commit messages.
- **Testing:** Test thoroughly in a development environment before pushing to production. Use test cases that cover both typical and edge-case scenarios.
- **Documentation:** Document all customizations and code changes, including configuration details and scripts.

- **Performance Optimization:** Avoid excessive database calls and leverage batch processing for large data sets.
- **Error Handling:** Implement robust error handling, especially for transaction-related customizations.

9. Debugging and Troubleshooting Tips

- **Logging:** Enable detailed logging for transactions, approvals, and business rules.
- **Console Output:** Monitor the console output in your IDE or server logs for runtime errors.
- **Database Checks:** Verify that all required data is present in the database, and query key tables directly for troubleshooting.

10. Deployment and Testing

- **Build and Deploy:** Use build scripts to package and deploy your application to a staging environment.
- **User Acceptance Testing (UAT):** Involve end-users in testing to validate that changes meet business requirements.
- **Production Release:** Coordinate with the system administrator to deploy changes to production during maintenance windows.

Additional Resources

- **CGI Advantage Documentation:** Refer to the official CGI Advantage documentation for in-depth guides.
- **Community and Support:** Reach out to the CGI user community or CGI support for troubleshooting and advanced assistance.
- **Code Repositories:** Maintain your code in a central repository for version control and team collaboration.

Summary

This crash course introduces developers to key aspects of CGI Advantage 4, covering architecture, customization, common workflows, and best practices. By following these guidelines, developers can effectively extend and maintain applications in CGI Advantage 4 to meet organizational needs.

Comprehensive Inventory for End-to-End Testing of CGI Advantage 4 - FIN and HRM Modules

Overview

This document outlines the core tables and required data elements for conducting an end-to-end test of CGI Advantage 4’s baseline functionality within the **Financial Management (FIN)** and **Human Resources Management (HRM)** modules. This setup assumes that all modules are configured with standard settings.

1. Core Tables and Data for Financial Management (FIN) Testing

The **Financial Management** module includes General Ledger, Accounts Payable, Accounts Receivable, Budgeting, and other essential financial functions. Below are the primary tables and their required data.

General Ledger (GL) Tables

Table Name	Description and Required Data
GL_ACCOUNT	Stores account codes, descriptions, and account types (e.g., revenue, expense).
GL_JOURNAL	Records journal entries, including transaction IDs, account codes, amounts, and dates.
GL_BALANCES	Holds balance information by account, fund, and period. Essential for testing period-end processing.

Accounts Payable (AP) Tables

Table Name	Description and Required Data
AP_INVOICE	Stores details of vendor invoices, including vendor ID, amount, invoice date, and due date.
AP_PAYMENT	Records payments made to vendors, with data on payment amount, date, and linked invoices.
AP_VENDOR	Vendor profile table with data on vendor name, contact info, and payment terms.
AP_VOUCHER	Vouchers for approved invoices, including vendor ID, amount, and linked invoice data.

Accounts Receivable (AR) Tables

Table Name	Description and Required Data
AR_INVOICE	Details customer invoices, including customer ID, amount, due date, and status.
AR_PAYMENT	Stores received payments, linking payment details to specific invoices and customers.
AR_CASH_RECEIPT	Cash receipt data for payments received, including payment method and receipt number.
CUSTOMER_PROFILE	Customer profile details, including name, billing address, and contact information.

Budgeting Tables

Table Name	Description and Required Data
BUDGET_STRUCTURE	Budget structure and hierarchy definitions, including budget codes and fund allocations.
BUDGET_ENTRY	Records budget entries, including original budget, adjustments, and allocations.
BUDGET_CONTROL	Budget control limits and rules, ensuring spending stays within budgeted amounts.

Procurement and Vendor Management Tables

Table Name	Description and Required Data
VENDOR_PROFILE	Vendor profile details, similar to AP_VENDOR, but used across multiple modules.
PURCHASE_ORDER	Stores purchase order data, including PO number, vendor, item details, and approval status.
CONTRACT	Holds contract details, linking vendors to terms, amounts, and related POs.

2. Core Tables and Data for Human Resources Management (HRM) Testing

The **HRM module** manages employee data, payroll, benefits, and timekeeping. Below are the core tables and necessary data.

Employee Profile and Address Tables

Table Name	Description and Required Data
EMPLOYEE_PROFILE	Basic employee data including employee ID, name, department, job title, and hire date.
EMPLOYEE_ADDRESS	Stores employee addresses, including address type (home, work) and contact info.

Payroll Tables

Table Name	Description and Required Data
PAYROLL_RUN	Tracks payroll runs, including pay period start and end dates, total payroll amount, and status.
PAY_CALCULATION	Stores calculated pay details per employee, such as gross pay, deductions, and net pay.
TAX_TABLE	Tax withholding rates and brackets by jurisdiction and employee status.

Benefits and Enrollment Tables

Table Name	Description and Required Data
BENEFIT_PLAN	Details on available benefits plans, including coverage types and premium amounts.
BENEFIT_ENROLLMENT	Enrollment records for employees, specifying benefits chosen, coverage levels, and effective dates.

Time and Attendance Tables

Table Name	Description and Required Data
TIME_ENTRY	Time entries for employees, tracking dates, hours worked, and time codes (e.g., REG, SICK, VAC).
LEAVE_BALANCE	Leave balances per employee, showing accrued and used leave by type (e.g., vacation, sick).

3. Supporting and Common Tables for FIN and HRM Modules

User and Security Tables

Table Name	Description and Required Data
USER_PROFILE	User login information, including username, role, and department.
ROLE_ASSIGNMENT	Maps users to roles, defining access permissions across modules.
WORKFLOW_CONFIG	Workflow rules for approvals, specifying approver levels and routing paths for various transactions.

Fiscal Period and Budget Period Tables

Table Name	Description and Required Data
FISCAL_PERIOD	Defines fiscal periods, including start and end dates for each accounting period.
BUDGET_PERIOD	Tracks the budget periods for use in budget and financial transactions.

4. Required Data for Testing

For a comprehensive end-to-end test, populate the following data:

Financial Management Data

- **Accounts and Budgets:** Complete account hierarchy, with active accounts in **GL_ACCOUNT** and linked budgets in **BUDGET_ENTRY**.
- **Vendors and Invoices:** Create vendors in **VENDOR_PROFILE** and populate with sample

invoices in **AP_INVOICE** and payments in **AP_PAYMENT**.

- **Customer and Receivables:** Populate customers in **CUSTOMER_PROFILE**, with sample receivables in **AR_INVOICE** and payments in **AR_PAYMENT**.

Human Resources Data

- **Employees and Payroll:** Populate employee profiles in **EMPLOYEE_PROFILE** and assign payroll details in **PAY_CALCULATION**.
- **Benefits:** Create benefits plans in **BENEFIT_PLAN** and enroll employees in **BENEFIT_ENROLLMENT**.
- **Time Entries:** Populate **TIME_ENTRY** with sample work hours and time-off entries, and set leave balances in **LEAVE_BALANCE**.

5. Testing Scenarios

Use the following testing scenarios for a full functional review:

General Ledger Testing

1. **Post Journal Entries:** Test journal entry creation, editing, and posting in **GL_JOURNAL**.
2. **Verify Balances:** Check updated balances in **GL_BALANCES** after transactions.

Accounts Payable and Receivable Testing

1. **Create AP Invoice:** Generate invoices in **AP_INVOICE** and process payments in **AP_PAYMENT**.
2. **Generate AR Invoice:** Create receivable entries in **AR_INVOICE** and test payments in **AR_PAYMENT**.

Payroll and Benefits Testing

1. **Run Payroll:** Execute payroll in **PAYROLL_RUN** and verify calculations in **PAY_CALCULATION**.
2. **Enroll Benefits:** Test benefit enrollment and verify deductions in payroll.

Budget Testing

1. **Set Budget Control:** Test budget constraints by entering data in **BUDGET_CONTROL** and attempting transactions that exceed limits.
2. **Budget Adjustments:** Make budget adjustments in **BUDGET_ENTRY** and verify updates.

Summary

This inventory provides the foundational tables, data, and test scenarios necessary for a comprehensive end-to-end test of CGI Advantage 4's baseline functionality in the Financial Management and Human Resources Management modules. Ensure all data is populated as specified and use the test scenarios to validate field, function, and feature performance across modules.